Long Live the Patient King

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Abstract

I propose a simple model of economic growth and extractive taxation. A citizen and a ruler interact over time. In every period, the ruler announces a tax rate, which determines the current distribution of output between the two players. The citizen then chooses whether to accept the proposed distribution or leave the game and secure a fixed outside option. If the citizen stays in the game, they then choose a level of *innovation effort*, which determines the future level of productivity. I define a *social contract* as an equilibrium of the repeated game. A social contract is *extractive* at a given history if it leaves the citizen indifferent between the on-path continuation and the outside option; and it is *non-extractive* otherwise.

I show that all Pareto efficient and renegotiation proof social contracts feature a *dual transition*. As time passes, society endogenously moves from a regime of slow innovation and low growth — where extractive social contracts are implementable — to a regime of fast innovation and high growth — where they are not. These findings suggest that modern rulers may be less extractive than ancient ones not (only) because constitutions constrain them, but (also) because citizens' work has become too productive to be discouraged. Using cross-country data from 1789 onward, I present empirical evidence consistent with this historical reading.

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1 Introduction

As argued by Weingast [1995], "the fundamental political dilemma of an economic system" is that "a government strong enough to protect property rights and enforce contracts is also strong enough to confiscate the wealth of its citizens," thereby deterring investments and growth (North and Weingast [1989]). However, history shows that the dilemma can be resolved. Over the very long run, the world has become both more prosperous and less despotically governed.

North and Weingast [1989] provide a possible solution to this puzzle that emphasizes the role played by constitutions and formal institutions more in general. In this view, constitutions are devices that force rulers to commit ex ante to non-extractive plans they would otherwise renegotiate ex post (Fleck and Hanssen [2006]). A related and influential strand of literature describes these commitments as the "fundamental" cause of growth and human progress (e.g., Acemoglu et al. [2001, 2005a]).

Yet, a society's successful transition out of an extractive regime might reflect a deeper change in the strategic interaction between citizens and rulers, of which the signing of a new constitution could be just a symbol or a symptom. Moreover, constitutions and formal systems of checks and balances must often be self-enforcing and robust to players' incentives to renegotiate them.

This article describes how societies can escape extraction and achieve sustained growth when neither rulers nor citizens have commitment power, and when institutional change must be robust to renegotiation.

I build a simple model where a citizen and a ruler repeatedly interact over an infinite horizon. The citizen owns the means of production. They possess a stock of capital which in every period produces some output according to a time-varying technology. The ruler has political power. They can tax the citizen's output at their discretion and freely consume the collected tax revenues.

The stage game is as follows. At the beginning of each period, the ruler announces a tax rate. The citizen observes it, and makes two subsequent decisions. They first

decide whether to quit the game or not. If the citizen quits the game, they secure a fixed outside option, and the ruler's payoff is zero forever thereafter. If the citizen stays in the game, they then choose an innovation effort, which is either low or high. High effort is costlier, but it increases the per-period growth rate of the economy and thus the level of output in the future. The ruler maximizes life-time consumption. The citizen maximizes life-time consumption net of the cost of effort.

In this model, the economy grows at a fast rate in every period (a situation I refer to as sustained growth) if and only if two conditions are satisfied. The first condition is immediate: the citizen must not quit the game. In this sense, the existence of a citizen-ruler *continued* relationship (which I then define as a social contract) is a necessary condition for growth. This feature of the model captures the idea that if the citizen quits the game, they lose access to their capital because — in the absence of a ruler — property rights are not enforced. The second, less immediate condition is that the citizen exerts high effort in every period.

In this regard, the main obstacle to growth is that any return to the citizen's effort in period t is realized only in period $t+1$, when the ruler can always expropriate it by increasing the tax rate. Anticipating this risk, the citizen may have no incentives to innovate — exactly as noted by North and Weingast [1989] and Weingast [1995].

When this happens in equilibrium, the social contract is *extractive*. The ruler raises taxes in every period up to the point in which the citizen's participation constraint binds. In turn, the citizen never exerts high effort and growth is minimized. When players are myopic, this is the unique possible equilibrium at every history.

By contrast, I show that when players are forward-looking, sustained growth and low rent extraction are not only possible, but also inevitable (up to a simple refinement on the equilibrium concept). On the path of every Pareto efficient and renegotiation proof subgame perfect equilibrium (or social contract), the simple society described in my model overcomes the "fundamental dilemma" of Weingast [1995] by means of a system of repeated and self-enforcing *promises*. In every period, the ruler promises low taxes

tomorrow in exchange for the citizen's effort today. Crucially, in the next period, the ruler has an incentive to keep their promise as they want to induce further growth. Put simply, the faster growth of the pie compensates the ruler for their reduced share. In this sense, non-extractive social contracts appear as *relational contracts* (e.g., Levin [2003]), that is, agreements sustained by the fact that players value the future dividends of their relationship.

However, the patient ruler must live long enough. The citizen has to wait some periods before the fast growth path is in the ruler's interest. At the beginning of the game, the citizen never exerts high effort and extractive social contracts are implementable. After a certain calendar period is reached, the political-economic equilibrium changes. Society experiences a *dual transition*: the citizen exerts high effort, growth takes off, and the ruler reduces the tax rate accordingly.[1] From that period onwards, every social contract is non-extractive as it must leave the citizen with a positive surplus.

On the path of these equilibria, the citizen's surplus is necessarily larger than the cost of effort, so that their net consumption flow strictly increases as a consequence of the ruler's tax concession. This is a consequence of the dynamic nature of the model. The citizen advances the cost of effort one period before being repaid for it. Each period of high effort can then be seen as a one-period loan from the citizen to the ruler, so that its repayment must cover the principal (the cost of effort) *and* the interest, which the citizen captures as a profit. For this reason, the citizen is necessarily a strict beneficiary of sustained growth, and not merely reimbursed for starting it.

The engine of the dual transition is the accumulation of knowledge. The return to the citizen's effort is proportional to the level of productivity already attained, while its cost is not. In this sense, I say that technological progress features *intertemporal complementarity*. The citizen's effort in period t increases the returns to further effort in period $t+1$. It follows that, as productivity rises, extraction becomes increasingly costly for the ruler, until it is no longer optimal.

[1] A famous parallel concept is the *dual revolution* of Hobsbawm [1962], which describes the emergence of the modern world as a joint consequence of the French Revolution and of the English industrial revolution.

From a social welfare perspective, however, the transition arrives too late. A benevolent social planner would dictate the citizen's effort as soon as the growth it generates is sufficient to cover its cost. In a decentralized equilibrium, this is not enough. The ruler must reward the citizen's effort with lower taxes, and is willing to do so only if the faster growth of the pie also compensates them for their smaller share. Growth must then be large enough to compensate *both* players, which requires a (weakly) higher level of productivity. This delay vanishes when players are patient enough, as the citizen's effort becomes both cheaper and more appealing for the ruler.

Even if delayed, the transition is inevitable. The inevitability of non-extractive social contracts implies that players can be successfully deterred from unilateral, selfish deviations. The ruler can be deterred from raising the tax rate once the citizen has already exerted high effort, and the citizen can be deterred from saving the cost of effort once the ruler has reduced rent extraction. To deter such deviations, social contracts prescribe punishments.

By subgame perfection, punishments must be itself equilibria of the off-path continuation. By the adopted notion of renegotiation proofness, such equilibria must be Pareto undominated. That is, following a deviation, at the beginning of every subsequent period, there must exist no other equilibrium which would make both players strictly better off. The intuition behind this refinement is that if such alternative Pareto dominating equilibrium existed, no player would oppose a renegotiation of the punishment.

A direct consequence of renegotiation proofness is that off-path continuations cannot prescribe a permanently slower growth path whenever faster growth would be both socially efficient and incentive compatible. Unlike classic work on folk theorems, players' cooperation can then be sustained in equilibrium for any discount factor.

To further illustrate the off-path properties of non-extractive social contracts, I focus on two particular contracts: the citizen- and ruler-preferred contracts. These two contracts correspond to the two boundaries of the set of renegotiation proof social contracts (which corresponds to the Pareto frontier). Under the ruler-preferred (respectively, citizen-preferred) contract, the citizen's (respectively, ruler's) value function is at its lower

bound at every history. Off-path, the two contracts serve as each other's punishment. A deviation by the ruler triggers one period of low effort followed by the citizen-preferred contract, and a deviation by the citizen triggers the ruler-preferred one. The harshest credible punishment is thus not to permanently shrink the pie, but to hand the entire surplus of the relationship over to the victim of the deviation.

The historical reading of these results is that modern rulers are less extractive than ancient ones not (only) because constitutions constrain them, but (also) because citizens' work has become so productive that discouraging it through extraction is no longer in the rulers' own interest. In this view, the concession of constitutions reflects a deeper transformation of the ruler's own incentives, which is ultimately triggered by growth.

In the second part of the paper, I test the main predictions of the model using cross-country data from 1789 to 2024. I identify 65 *dual transitions*, i.e., episodes in which a permanent increase in the constraints faced by the executive is accompanied by a faster growth rate of GDP. I show that, after dual transitions, rulers appropriate a smaller share of output. However, output grows so much that, in most cases, they are not worse off in the long run, consistent with the dual transition being a Pareto improvement, rather than a forced transfer of power.

The remainder of the paper is organized as follows. In the next section, I briefly summarize the related literature. Section 3 presents the model. Section 4 shows some preliminary results. Section 5 introduces renegotiation proofness and presents the main result. Section 6 describes the two players' preferred contracts. Section 7 discusses welfare and the model's main assumptions. Section 8 presents empirical evidence supporting the main theoretical predictions. Section 9 concludes. Proofs and additional details are relegated to the Appendix.

2 Literature Review

I contribute to two main literatures. The first is the literature on the theory of the state. In a recent survey of this literature, Heldring [2025] organizes the main existing contributions along a "fundamental dichotomy" between two distinct streams of research.

According to the *cooperative* tradition, the state emerges as a solution to collective action problems. Individuals voluntarily give up part of their freedom to a sovereign entity in exchange for the provision of public goods, such as the enforcement of property rights or public safety (Hobbes [1651]; Locke [1689]; Rousseau [1762]; Nozick [1974]).[2] It follows that, as the provision of public goods affects productivity, a government's activity has a direct impact on growth (Barro [1990]).

By contrast, the *coercive* theories of the state emphasize the role of force. In this view, the state is an instrument controlled by "one class for oppressing another" (Marx and Engels [1848]). In Olson [1993]'s famous framework, states formed as warrior elites found it profitable to settle, claim their coercive power over a territory, and extract rents from its inhabitants on a regular basis.[3]

Recent empirical research has provided evidence consistent with both traditions (e.g., Sánchez de la Sierra [2020], Mayshar et al. [2022], and Allen et al. [2023]). My model is agnostic on the origin of the state, although it combines ideas from both traditions (social contracts are necessary for the enforcement of property rights and growth, as in the cooperative view; while the ruler can expropriate resources from the economy, as in the coercive view). I contribute to this literature by focusing how the ruler *exercises* — rather than *takes* — power, and thus by providing a theory of how states evolve over time.[4]

[2]For recent formal contributions on the social contract as a self-enforcing agreement, see, e.g., Tyson [2020, 2023].

[3]A key implication is that because these "stationary bandits" expect to remain in power, they have an incentive to limit extraction in order to preserve the productive capacity of the economy. In my model, the ruler faces similar incentives. However, because of intertemporal complementarity, they reduce rent extraction *only* after the economy has reached a certain level of productivity.

[4]For this reason, my paper is most closely related to Acemoglu and Robinson [2023] (see also Acemoglu and Robinson [2019]). They model a dynamic contest between citizens and rulers, and characterize the equilibrium type of state in terms of the balance of power between players. By contrast, my model emphasizes the transition between two regimes, thereby focusing more on the transitional dynamics than on the steady state.

This paper is also related to the literature on democratization, institutional change, and growth.[5] Economic development can affect both the *demand* for and the *supply* of democracy.

On the demand side, development may empower under-represented social groups in their struggle for political rights. For instance, Lipset [1959] famously argues that since richer societies are more educated and have a larger middle class, growth can explain support for democracy. According to Moore [1966], the start of growth and trade empowered and enriched a social class (the bourgeoisie) whose income did not depend on the crown, and which could therefore oppose absolutism. According to Rueschemeyer et al. [1992], as industrialization brought workers together in factories and cities, this allowed them to organize in unions and parties, and thus to push for obtaining voting rights.

On the supply side, this literature asks why did powerful elites "extend the franchise". The leading answer is that elites made concessions because they were forced to. For instance, Acemoglu and Robinson [2000] argue that the elite extends the franchise when it faces the threat of a revolution. In their setting, the threat of revolt is transitory, but it can be prevented only through transfers that extend into the future. A promise to redistribute tomorrow, however, would not be credible because once the threat has passed, the elite would withdraw the concession. Therefore, the only possible way for the elite to prevent a revolution is by conceding political rights. In this framework, democracy emerges as elites lack commitment power (see also Acemoglu and Robinson [2001, 2006]). In Conley and Temimi [2001], elites extend the suffrage to eliminate the costs of social conflict. Aidt and Franck [2015] provide evidence that the threat of riots increased support for reform in 19th-century Britain. In this tradition, growth can affect the balance of power between different social groups and thereby influence institutions.

My model departs from this tradition in that it does not need the threat of revolt to explain institutional change. It is thus closer in spirit to a related literature in which

[5] A related empirical literature estimates the *reverse* effect, i.e., the effect of democracy on growth (e.g., Barro [1996]; Papaioannou and Siourounis [2008]; Acemoglu et al. [2019]).

elites limit their power *voluntarily*. For instance, Lizzeri and Persico [2004] argue that the elite may extend the franchise to move government spending away from targeted transfers (pork-barrel politics) toward productive public goods. Relatedly, Galor and Moav [2006] argue that elites may voluntarily give up their privileges and support mass education as *their* physical capital is more productive when workers are more skilled. Similarly, Cervellati et al. [2025] shows that the onset of sustained growth hastens the accumulation of human capital, thereby reducing distributive conflict and reducing elites' resistance to democracy.

My model contributes to the literature on socially efficient political change by providing a parsimonious explanation in which the only driving force behind the transfer of resources from extractive rulers to productive citizens is economic growth.

3 Model

A citizen and a ruler repeatedly interact over time. The citizen owns a stock of capital of value $k > 0$, which in every period produces output $y_t = A_t k$, where $A_t > 0$ is the level of productivity (or the state) at the beginning of period t.

Time is discrete and runs over an infinite horizon. In every period $t = 1, 2, \ldots$, the citizen and the ruler play the following stage game.

The Stage Game The ruler announces a tax rate, $\tau_t \in [0, 1]$. The citizen observes it, and makes two decisions in sequence. First, they decide whether to stay in the game or not: $p_t \in \{S, Q\}$. If the citizen quits ($p_t = Q$), the game ends. The citizen's payoff is equal to the outside option, $\bar{u} > 0$, and the ruler's payoff is zero. If instead the citizen stays ($p_t = S$), they collect the output left by the ruler, and the game continues. Second, the citizen chooses an innovation effort, which is either low or high: $e_t \in \{l, h\}$.

Innovation effort does not affect current output. It determines the growth rate of the economy: $A_{t+1} = g_{e_t} A_t$, where $g_h > g_l > 1$, and the initial condition, $A_1 > 0$, is given.

At the end of each period, output is produced, players' per-period payoffs are realized, and the state variable updates.

Payoffs Suppose the history of the game reaches period t. Let $c_h > c_l \geq 0$ be the citizen's cost for high and low effort, respectively. For simplicity, and without consequences on the qualitative nature of the results, I normalize $c_l = 0$. The citizen's per-period level of consumption (net of the cost of effort) is $x_{C,t} = (1 - \tau_t)\, y_t - c_h\, \mathbf{1}\{e_t = h\}$. The ruler's consumption level is $x_{R,t} = \tau_t y_t$.

Both players maximize the discounted sum of their life-time consumption flows. The common discount factor is $\delta \in [0, 1)$. I assume $\delta g_h < 1$, so that discounted payoffs are finite.

Figure 1 summarizes the stage game.

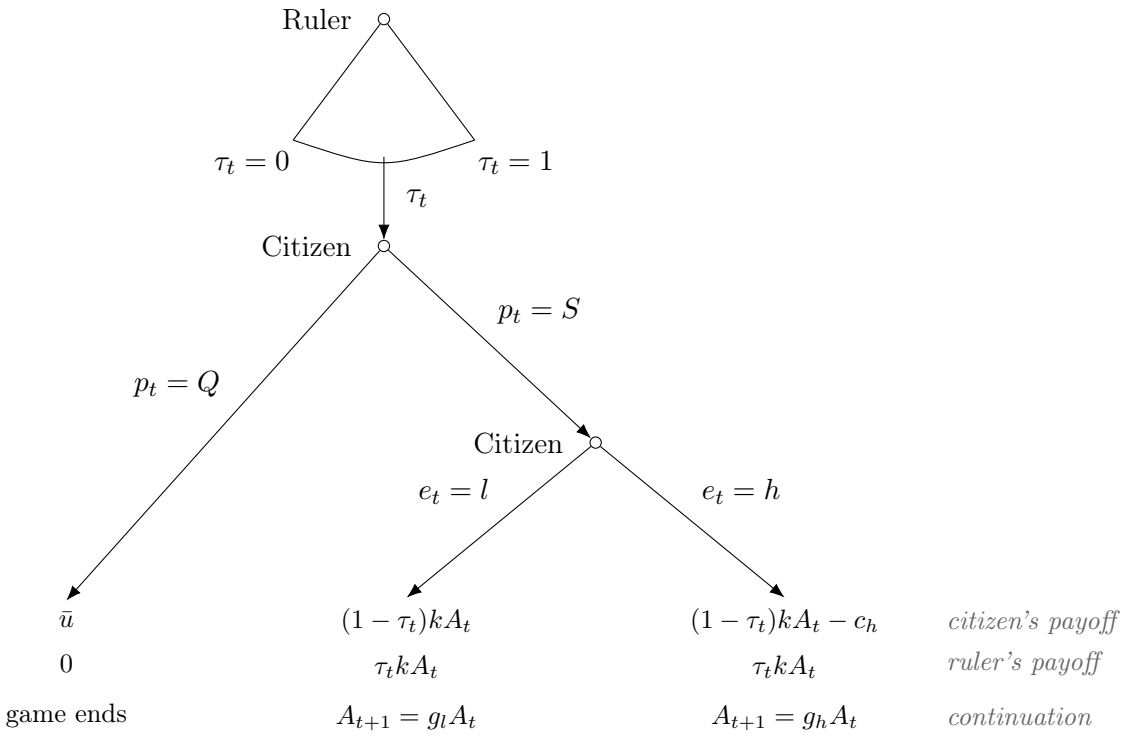


Figure 1: **The stage game in period** t.

Notation and Equilibrium Concept Let $H_t := \{\tau_r, e_r\}_{r=1}^{t-1}$ denote the history of the game up to the beginning of period t,[6] and let $A_t(H_t)$ denote the level of productivity implied by history H_t through the sequence $\{e_r\}_{r=1}^{t-1}$. For a strategy profile σ and at a given history H_t, I denote by $U_C(\sigma|H_t)$ and $U_R(\sigma|H_t)$ the citizen's and the ruler's life-time payoffs, respectively. Throughout the remainder of the paper, I often omit time subscripts whenever not strictly necessary.

[6]The sequence $\{p_r\}_{r=1}^{t-1}$ is omitted from the history, as $p_r = S$ at every history reached by the game.

Social contracts (i.e., agreements between rulers and citizens) must be self-enforcing, as players naturally lack commitment power. For this reason, I posit the following definition:

Definition 1 (Social Contracts)**.** *A social contract is a profile of strategies which constitutes a subgame perfect equilibrium of the game.*

Social contracts differ in how much output the ruler leaves to the citizen. I classify them accordingly:[7]

Definition 2 (Extractive and Non-Extractive Social Contracts)**.** *A social contract σ is extractive at a given history H if the citizen's continuation value is equal to the outside option, $U_C(\sigma|H) = \bar{u}$. If instead $U_C(\sigma|H) > \bar{u}$, the social contract σ is non-extractive.*

4 Preliminary Results: Extractive Contracts

4.1 The Myopic Benchmark

To begin with, suppose players are myopic: $\delta = 0$. Players only care about their current period payoff. The citizen's effort raises output only in the future. Therefore, a myopic citizen never exerts high effort. Anticipating this, the ruler's optimal strategy is to maximize rent extraction.

In equilibrium, the citizen's participation constraint binds. The ruler can never afford to let the citizen quit the game, since that would leave the ruler with a payoff of zero. This endogenously constrains the ruler's extractive power. The citizen does not quit the game if $(1-\tau)\,kA \geq \bar{u}$. Therefore, the ruler's optimal tax rate is

$$\tau = 1 - \frac{\bar{u}}{kA} =: \tau^m(A)\ . \tag{1}$$

Therefore, when $\delta = 0$, every social contract is extractive at every on-path history.

[7] The classification omits the case in which $U_C(\sigma|H) < \bar{u}$, as this case can never emerge in equilibrium (Proposition 1).

4.2 The Extractive Social Contract

Can extractive social contracts emerge also when players are forward-looking ($\delta > 0$)? In such contracts, the citizen's continuation value is $\bar{u}$ at every on-path history, so their flow payoff, x_C, satisfies $\bar{u} = x_C + \delta\bar{u}$, i.e., $x_C = (1-\delta)\,\bar{u} =: \phi$. To leave the citizen a flow payoff of ϕ (and, thereby, a constant continuation value of $\bar{u}$), the tax rate announced by the ruler satisfies[8]

$$(1-\tau)\,Ak = \phi \Rightarrow \tau = 1 - \frac{\phi}{kA} =: \tau^e(A)\ . \tag{2}$$

I now show that such a tax rate is implementable in equilibrium. Let σ^e denote the extractive social contract, defined as follows. At every state A, the ruler plays $\tau^e(A)$. The citizen stays ($p = S$) and exerts low effort ($e = l$) if and only if $\tau \leq \tau^e(A)$; otherwise, they quit ($p = Q$).

Proposition 1 (The Extractive Lower Bounds)**.** *The profile of strategies σ^e is a social contract. Under this contract, the ruler's value function is $U_R^e(A) := \frac{kA}{1-\delta g_l} - \bar{u} > 0$, and the citizen's value function is $\bar{u}$. Then, σ^e is extractive at every on-path history. Moreover, for any social contract σ and at every (on- or off-path) history H*

$$U_R(\sigma|H) \ \geq\ U_R^e(A(H)) \qquad \text{and} \qquad U_C(\sigma|H) \ \geq\ \bar{u}\,. \tag{3}$$

Proposition 1 contains two results. First, it provides an illustration of the prominent argument according to which non-extractive institutions are a "fundamental cause" of long run growth (Acemoglu et al. [2001, 2005a]).

The citizen bears the full cost of effort but only internalizes a fraction of the additional output it generates. Therefore, their incentives to exert high effort — which increases the growth rate of the economy — depend on the rent extraction rate implemented by the ruler in the future. It follows that the citizen exerts high effort if and only if they *expect* rent extraction in the future to be low enough. Under an extractive contract, these expectations are not fulfilled. As a result, sustained growth never takes off.

[8] I assume $kA_1 \geq \phi$, so that Equation (2) is always lower or equal than one.

Second, the proposition provides a lower bound on what each player can obtain in any social contract. The citizen can always quit and secure the outside option, $\bar{u}$. The ruler can always set $\tau^e(A)$ and — since σ^e is itself a social contract — secure $U_R^e(A)$. It follows that a transition toward a non-extractive contract is possible only if it leaves the ruler at least $U_R^e(A)$, i.e., if the ruler is never made worse off by reducing rent extraction.

If $\delta = 0$, the tax rate of the extractive social contract (Equation (2)) and the equilibrium tax rate of the one-shot game (Equation (1)) coincide. If $\delta > 0$, instead, $\tau^e(A) > \tau^m(A)$. The reason is that the binding constraint for the ruler is the citizen's *life-time* participation constraint. A myopic citizen must be paid their entire outside option in every period. A patient citizen, instead, accepts a strictly smaller flow payoff in every period, $\phi = (1-\delta)\bar{u} < \bar{u}$, because they anticipate the value of the future. In this sense, the citizen's patience works against them.

Proposition 1 has two further implications. First,

Remark 1. *Under any social contract, the citizen never quits; $p_t = S$ at every on-path history.*

As discussed in the Introduction, the existence of a relational contract between the citizen and the ruler is a necessary (but not sufficient) condition for sustained growth.

Second,

Remark 2. *The social contract σ^e is the (essentially) unique contract which is extractive at every on-path history.*

The argument supporting this remark is simple. If a social contract is extractive at every on-path history, the citizen's continuation value is equal to $\bar{u}$ along the whole path. This univocally pins down the equilibrium tax rate, $\tau^e(A_t)$ — as shown above. The contract is therefore essentially unique because while on-path play is univocally characterized, different contracts may support this equilibrium play through different off-path punishments.

Figure 2 plots the paths of the equilibrium tax rate and of the two players' value functions under the extractive social contract.

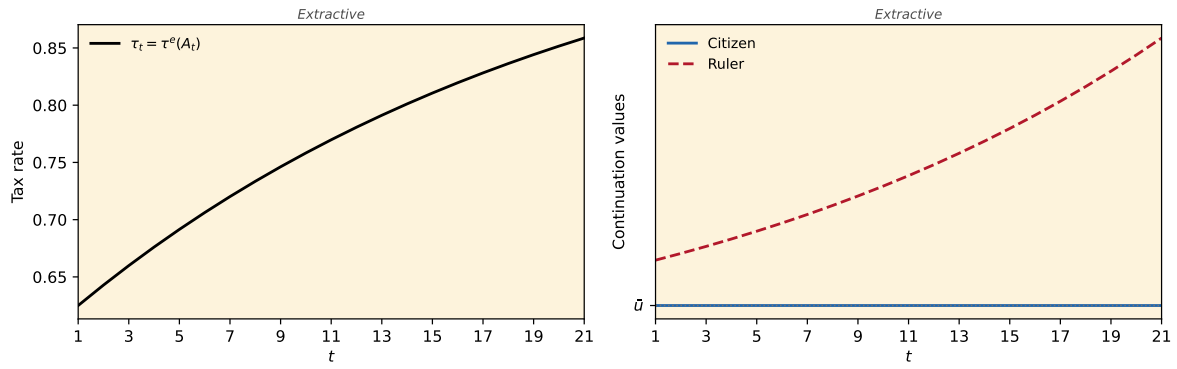


Figure 2: **The extractive social contract**. Equilibrium tax rate (left), and players' continuation values (right).
Note: Parameters: $\delta = 0.7$, $k = 1$, $\bar{u} = 1$, $c_h = 0.5$, $g_l = 1.05$, $g_h = 1.15$, $A_1 = 0.8$.

To sum up, while extraction is the unique possible outcome if players are short-lived, it can also arise when they are not. Patience, therefore, does not reduce extraction *by itself*.

5 Renegotiation Proof Social Contracts

Patience, however, allows players to make *promises*, which are necessary to sustain growth because the citizen's effort in period t can only be repaid by lower taxes in period $t+1$.

5.1 The Social Value of Growth and the First Best Problem

To illustrate the societal value of such promises, suppose the citizen exerts high effort in every period. The present value of the sum of the two players' payoffs is

$$\Pi_h(A) := \sum_{t\geq 0} \delta^t \left(kAg_h^t - c_h\right) = \frac{kA}{1-\delta g_h} - \frac{c_h}{1-\delta}. \tag{4}$$

If instead the citizen always exerts low effort, social welfare is equal to

$$\Pi_l(A) := \sum_{t\geq 0} \delta^t kAg_l^t = \frac{kA}{1-\delta g_l}. \tag{5}$$

The net social surplus from sustained growth is

$$\Delta(A) := \Pi_h(A) - \Pi_l(A) = \frac{\delta(g_h - g_l)kA}{(1-\delta g_h)(1-\delta g_l)} - \frac{c_h}{1-\delta}, \tag{6}$$

which is linear increasing in A.

The reason why $\Delta(A)$ increases in A is the complementarity between existing and new technologies, or ideas. The *additional* output generated by the citizen's effort is equal to $kA(g_h - g_l)$. The benefit of effort is thus proportional to the level of productivity already attained, while its cost, c_h, is not. In this sense, I say that technological progress exhibits *intertemporal complementarity*. The return to discovering or implementing a new idea is proportional to the stock of ideas already in use (see Section 7). It follows that sustained growth can be optimal only once productivity is high enough.

To illustrate this point, consider the problem of a social planner who can dictate the citizen's effort to maximize social surplus, ignoring incentive compatibility. Notice that the tax rate affects only the distribution of surplus and not its level. I refer to this as the *social planner problem*, and to its solution as the *first best plan*.

Because of intertemporal complementarity, this problem is convex (see Appendix B). Then,

Lemma 1 (Growth Never Stops)**.** *In the social planner problem, if exerting high effort is optimal at state A, it is optimal at every state $A' \geq A$.*

By Lemma 1, once high effort starts, it never stops. The only remaining question is then *when* high effort should start. Postponing growth is suboptimal when the discounted surplus from starting high effort today rather than tomorrow exceeds its cost:

$$\delta\left[\Pi_h(Ag_h) - \Pi_h(Ag_l)\right] - c \;\geq\; 0 \qquad \Longleftrightarrow \qquad A \;\geq\; \hat{A} := \frac{c_h(1-\delta g_h)}{\delta k(g_h - g_l)}. \tag{7}$$

The first best plan therefore prescribes low effort until the state reaches an *efficiency* threshold, $\hat{A}$, and high effort forever thereafter.

5.2 Incentive Compatibility

Being socially desirable, however, does not make high effort incentive compatible. Consider a contract σ which prescribes the citizen's high effort at some history H. The tax rate τ has already been announced, so the citizen's current consumption flow is the same whether they exert high or low effort. Low effort therefore saves them the cost, c_h, today, and, by Proposition 1, still secures a continuation value of at least $\bar{u}$ from period $t+1$ onwards. High effort is thus incentive compatible only if $-c_h + \delta U_C(\sigma|H_{t+1}) \geq \delta\bar{u}$, that is, only if the contract promises the citizen a continuation value of at least

$$\omega := \bar{u} + \frac{c_h}{\delta} > \bar{u}\,. \tag{8}$$

The premium, $\frac{c_h}{\delta}$, is the *price* the ruler needs to pay to induce the citizen's high effort.

In equilibrium, this price must be affordable for the ruler. Suppose the citizen exerts high effort at state A, and that high effort continues in every subsequent period, as in the first best plan. Next period, the state is Ag_h, and the continuation surplus is $\Pi_h(Ag_h)$. The citizen must receive at least ω, and the ruler at least their lower bound, $U_R^e(Ag_h)$ (Proposition 1). Feasibility requires $\omega + U_R^e(Ag_h) \leq \Pi_h(Ag_h)$. Using $U_R^e(A) = \Pi_l(A) - \bar{u}$,

$$\Delta(Ag_h) \;\geq\; \frac{c_h}{\delta} \qquad \Longleftrightarrow \qquad A \;\geq\; \tilde{A} := \frac{c_h(1-\delta g_h)(1-\delta g_l)}{\delta^2(1-\delta)\,k\,g_h(g_h-g_l)}\,. \tag{9}$$

I refer to $\tilde{A}$ as the *incentive compatibility* threshold. When the state is above this threshold, the surplus created by permanent high effort is large enough to cover both the citizen's premium and the ruler's lower bound.

5.3 Equilibrium Refinement

In the remainder of the section, I focus on social contracts that are robust to renegotiation. As we will see, this property implies that social contracts are socially efficient both on and off the equlibrium path.

In a socially efficient social contract, high effort must be both worthwhile and re-

wardable. Accordingly, let

$$A^* := \max\left\{\hat{A},\ \tilde{A}\right\}, \tag{10}$$

and call *threshold rule* the plan which prescribes high effort if and only if $A \geq A^*$. Finally, let

$$T^* := \min\left\{t\ :\ A_1 g_l^{t-1} \geq A^*\right\}\ \geq\ 2 \tag{11}$$

denote the calendar period at which an economy growing at rate g_l reaches A^*, given the initial condition.[9] When $\tilde{A} > \hat{A}$, incentive compatibility postpones growth relative to the first best plan. I return to this inefficient delay in Section 7.

Social contracts also need to prescribe punishments for those who break their promises. By subgame perfection, punishments must be equilibria of the continuation (off-path) game themselves. However, not all self-enforcing punishments are equally deterrent. For instance, when entering a punishment phase in period t, after a deviation has occurred in period $t-1$, players may jointly abandon the previously agreed punishment in favor of another self-enforcing agreement that both prefer. The literature refers to equilibria sustained by such possibly implausible punishments as *not* renegotiation proof (see Farrell and Maskin [1989]; Levin [2003] or Acemoglu et al. [2008] for a political economy application).

Formally,

Definition 3 (Renegotiation Proof Social Contracts)**.** *A social contract σ is renegotiation proof (RP) if, at the beginning of every period, after any (on- or off-path) history H, there is no social contract of the continuation game, σ', such that $U_C(\sigma'|H) > U_C(\sigma|H)$ and $U_R(\sigma'|H) > U_R(\sigma|H)$.*

For any state A, let $\Sigma(A)$ denote the set of RP social contracts of the game with initial state A.

[9] I assume $A_1 < A^*$, and $A_1 g_l^{t-1} \neq \hat{A}$ for every t. The latter restriction excludes the special case in which society is indifferent between starting high effort at time T^* or T^*+1.

5.4 Equilibrium Properties

I now study renegotiation proof social contracts. My main result is:

Proposition 2 (The Dual Transition)**.** *The set of RP social contracts is non-empty, i.e.,* $\Sigma(A_1) \neq \emptyset$*. For every* $\sigma \in \Sigma(A_1)$*, on the equilibrium path, the following properties hold.*

(i) The growth rate of the economy is g_l *before* T^**, and* g_h *from* T^* *onwards.*

(ii) The citizen's continuation value, $U_C(\sigma|H_t)$*, is such that*

$$U_C(\sigma|H_t) \in \begin{cases} [\ \bar{u}\ ,\ \ \bar{u} + \delta^{T^*-t}\Delta(A_{T^*})\ ] & \text{if } t \leq T^*, \\ [\ \omega\ ,\ \ \bar{u} + \Delta(A_t)\ ] & \text{if } t > T^*, \end{cases} \tag{12}$$

and the equilibrium tax rate is pinned down by the citizen's Bellman equation:

$$(1-\tau_t)\,kA_t = U_C(\sigma|H_t) - \delta U_C(\sigma|H_{t+1}) + c_h\,\mathbf{1}\{t \geq T^*\}\ . \tag{13}$$

(iii) The contract is non-extractive at every on-path history with $t > T^*$*. As a consequence,* $\tau_t < \tau^e(A_t)$ *at some* $t > T^*$*.*

Sketch of the Proof The proof is presented in Appendix A. It proceeds in four steps. I start by formally defining the largest total surplus that equilibrium play can generate when the state is A, that is, $\bar{\Pi}(A)$ (see Definition 4, in the Appendix). Note that this is a second best measure of efficiency, as it need not coincide with the value of the first best plan.

The first step shows that $\bar{\Pi}(A)$ is attained by the threshold rule (Lemma 2). That is, it shows that, as in the first best plan, once high effort starts, it never stops. Below A^*, high effort is either inefficient (when $A < \hat{A}$) or cannot be rewarded by any social contract (when $A < \tilde{A}$).

The second step concerns the division of $\bar{\Pi}(A)$. The tax rate does not affect total surplus, but only how surplus is split. Hence, two contracts that prescribe the same path

of effort generate the same surplus, and differ only in its division. Lemma 3 shows that *every* division of $\bar{\Pi}(A)$ which respects the two lower bounds of Proposition 1 is attainable by a social contract. This step is important as it highlights the bite of renegotiation proofness. Without Lemma 3, a wasteful contract might be Pareto undominated simply because the efficient surplus could not be divided in a way that benefits both players.

The third step uses this intuition to connect renegotiation proofness with second best efficiency. Whenever a contract wastes surplus, there always exists one of the contracts of the second step that makes both players strictly better off, so that the first contract is not RP. Conversely, a contract that never wastes surplus leaves no room for Pareto efficient renegotiation. It follows that a social contract is RP *if and only if* it attains $\bar{\Pi}(A)$ at every history (Lemma 4). Therefore, any RP social contract prescribes the threshold rule.

The fourth step shows existence and proves the three equilibrium properties.

Proposition 2 describes the equilibrium path of institutional change. At the beginning of every on-path history, the economy grows slowly. After a certain calendar period is reached, economic growth accelerates permanently. The economic boom is followed by a permanent tax concession the ruler grants to the citizen.

In this regard, the size of the tax concession deserves a remark:

Remark 3. *At every $t > T^*$, the present value of the citizen's consumption plan along any RP social contract exceeds $\bar{u} + \frac{c_h}{1-\delta}$ (i.e., the outside option plus the full compensation for the cost of repeated effort) by at least $K := \frac{c_h}{\delta}$.*

Why can't the ruler simply compensate the cost of effort? The answer comes from the dynamic nature of the relationship. Sustained growth comes from the citizen's repeated effort. This dynamics can be seen as a sequence of one-period loans from the citizen to the ruler. In fact, the citizen pays the cost of effort in period t and is rewarded only in period $t+1$.

For the loan to be optimal for the lender, the repayment must cover the principal, c_h, plus the interest. The net interest per loan is $\frac{c_h}{\delta} - c_h = \frac{c_h(1-\delta)}{\delta}$. The *rent* K capitalizes this

perpetual interest, as the present value of infinitely many of such payments is exactly $\frac{c_h}{\delta}$. To summarize, because the growth process is dynamic, the citizen is a strict beneficiary of growth rather than merely reimbursed for producing it.

The ruler's trade-off is also intertemporal. Failing to repay the debt (i.e., raising the tax rate in period t when the citizen had exerted high effort in period $t-1$) would increase the ruler's current period payoff. However, the citizen might punish this behavior (see the proof of Lemma 3, in Appendix B, for an explicit construction of the punishments). Therefore, the ruler's restraint from eating more pie today is the price they have to pay in order to have a larger pie tomorrow.[10]

At higher levels of productivity, raising rent extraction above the level at which the citizen is willing to exert high effort implies a larger output loss for society. For this reason, once productivity has reached a certain threshold, the ruler also prefers a permanently lower share of a growing pie, rather than the whole of a slowly growing one.

Statement (ii) also shows that, while total surplus is necessarily maximized on the path of any RP social contract, its *distribution* is not univocally pinned down. In particular, Equation (12) characterizes the *equilibrium payoff set*, i.e., the set of points in the (t, U_C) space such that some RP social contract delivers the citizen a continuation value of U_C at time t (Lemmas 3 and 4). Given a sequence within this set, Equation (13) pins down the corresponding path of announced tax rates. Since the two players' value functions always sum up to the same value, the dynamics of the citizen's value function univocally characterizes the entire contract (on-path).

In equilibrium, the lower bound on the citizen's life-time payoff at a given history depends on the distance in time to the transition, T^*. Before the institutional transition, the floor is simply equal to the outside option. After the transition, the citizen's floor increases (see Remark 3). The ceilings are instead obtained as lower bounds on the ruler's life-time payoff.

The width of the equilibrium payoff set thus changes over time. In particular, very

[10] I return to players' incentive compatibility constraints when focusing on the two players' preferred contracts, in Section 6.

far from the transition, *every* renegotiation robust social contract is extractive. Figure 3 plots the equilibrium payoff set. Its two boundaries correspond to the players' preferred contracts, σ_C^* and σ_R^*. Section 6 characterizes them in detail. Every other RP social contract lies in between.

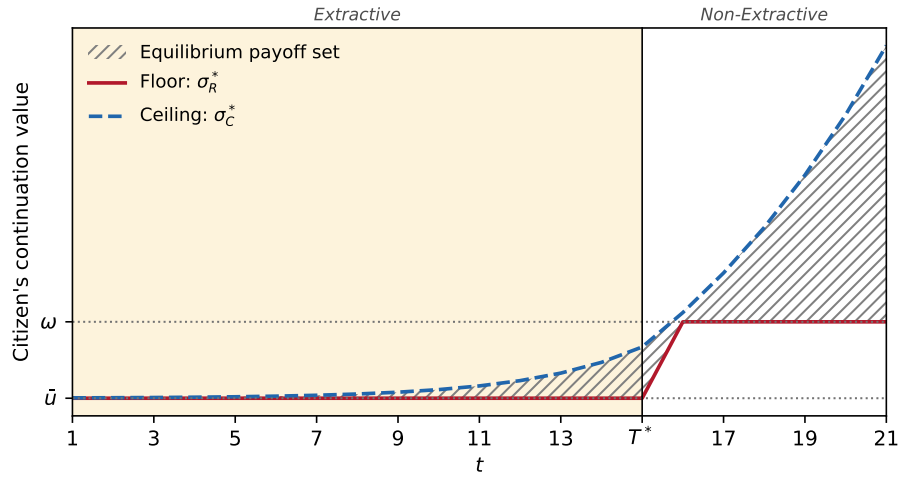


Figure 3: **The citizen's value function under any RP social contract**.
Note: Parameters as in Figure 2, which imply $\tilde{A} > \hat{A}$, $A^* \approx 1.53$, and $T^* = 15$.

Because any point in the equilibrium payoff set is attainable by a social contract, social welfare is maximized also off-path. Two further remarks follow. First,

Remark 4. *The extractive social contract is not renegotiation proof, i.e.,* $\sigma^e \notin \Sigma(A_1)$.

Under the extractive social contract, the growth rate of the economy is minimized. By contrast, Lemma 2 shows that sustained growth is achievable after T^*. Therefore, whenever the two players find themselves at a history whereby high effort would be Pareto efficient, no RP social contract can prescribe extraction. Players would renegotiate that.

Therefore,

Remark 5. *Consider any social contract that punishes some deviation by permanent reversion to* σ^e. *Then the contract is not renegotiation proof.*

In other words, a non-extractive, renegotiation robust social contract cannot be sustained by the threat of a permanent revert to the extractive equilibrium (i.e., by grim trigger strategies). What sustains it instead is the subject of the next section.

6 The Two Players' Preferred Contracts

Proposition 2 characterizes what every RP social contract must deliver. This section focuses on the two contracts at the boundaries of the equilibrium payoff set (Figure 3). Under the *ruler-preferred* contract, σ_R^*, the citizen's value function is equal to its lower bound at every on-path history. Under the *citizen-preferred* contract, σ_C^*, the ruler's value function is. By Statement (ii) of Proposition 2, each contract is essentially unique.[11]

Both contracts are constructed in detail in the proof of Lemma 3, in Appendix B. In what follows, I briefly summarize their on-path prescriptions. Off-path properties are discussed in Section 6.2.

On-path, both contracts prescribe the same actions for the citizen. The citizen exerts high effort if and only if $A \geq A^*$, and stays in the game if and only if $\tau \leq \tau^e(A)$. The two contracts differ only in the tax rate implemented by the ruler, and therefore in how the surplus generated by growth is split between the two players.

The ruler-preferred contract Under σ_R^*, the citizen's value function is equal to the lower bound. The ruler receives the remainder of the surplus:

$$U_C(\sigma_R^*|H_t) = \begin{cases} \bar{u} & \text{for } t \leq T^*, \\ \omega & \text{for } t > T^*, \end{cases} \qquad U_R(\sigma_R^*|H_t) = \bar{\Pi}(A_t) - U_C(\sigma_R^*|H_t).$$

The implied tax rates are

$$\tau_t = \tau^e(A_t) \quad \text{for } t \leq T^*, \qquad \tau_t = 1 - \frac{\phi + c_h/\delta}{kA_t} \quad \text{for } t > T^*.$$

Therefore, the ruler-preferred contract is extractive at every period $t \in [1, T^*]$, and non-extractive forever thereafter. Figure 4 plots the equilibrium tax rate, and the two players' value functions along this contract.

[11] See Remark 2 for a discussion on *essential* uniqueness.

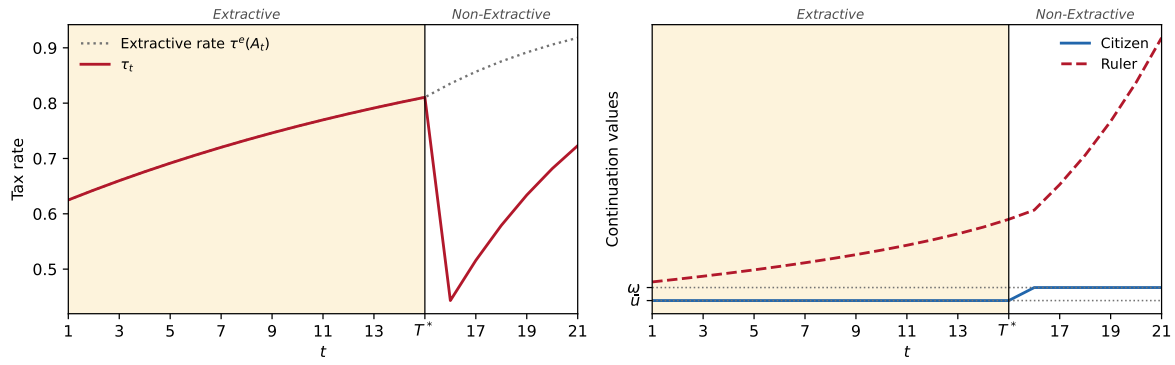


Figure 4: **The ruler-preferred contract**. Equilibrium tax rate (left), and players' continuation values (right).
Note: Parameters as in Figure 2, which imply $\tilde{A} > \hat{A}$, $A^* \approx 1.53$, and $T^* = 15$.

The citizen-preferred contract Under σ_C^*, the citizen's value function is equal to the upper bound. The ruler receives the remainder of the surplus:

$$U_C\left(\sigma_C^* | H_t\right) = \begin{cases} \bar{u} + \delta^{T^*-t}\Delta\left(A_{T^*}\right) & \text{for } t < T^*, \\ \bar{u} + \Delta\left(A_t\right) & \text{for } t \geq T^*, \end{cases} \qquad U_R\left(\sigma_C^* | H_t\right) = U_R^e\left(A_t\right).$$

The corresponding tax rates are

$$\tau_t = \tau^e\left(A_t\right) \quad \text{for } t < T^*, \qquad \tau_t = \tau^e\left(A_t\right) - \frac{\delta\left(g_h - g_l\right)}{1 - \delta g_l} \quad \text{for } t \geq T^*.$$

In this case, the citizen's rent (the difference between their value function and the outside option) is strictly positive at every period. Hence, σ_C^* is non-extractive at *every* on-path history. Figure 5 illustrates this social contract.

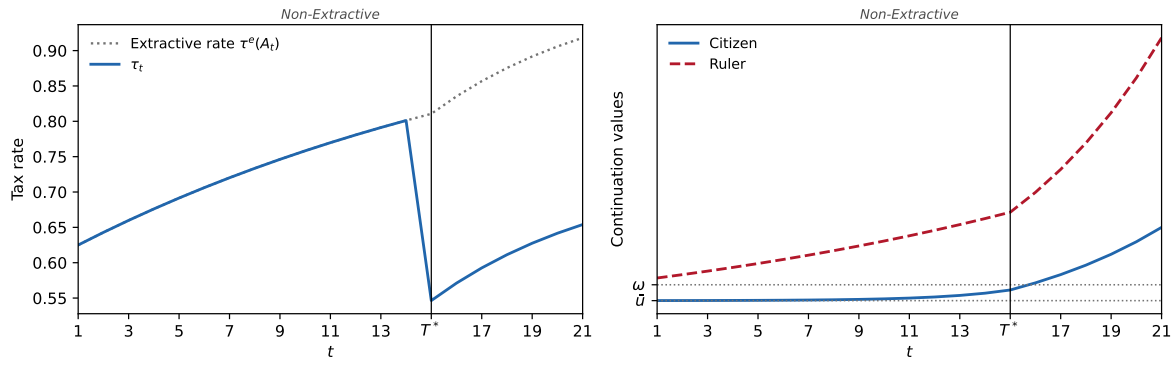


Figure 5: **The citizen-preferred contract**. Equilibrium tax rate (left), and players' continuation values (right).
Note: Parameters as in Figure 2, which imply $\tilde{A} > \hat{A}$, $A^* \approx 1.53$, and $T^* = 15$.

6.1 Comparing the Two Contracts: On-Path

Before the transition (for every $t < T^*$), the two contracts characterized in this section are observationally equivalent. They prescribe the same equilibrium tax rate, which also coincides with the tax rate prescribed by the extractive social contract derived in Proposition 1.

However, the citizen-preferred contract is always non-extractive, meaning that the citizen finds it always *strictly* optimal to stay in the game. This happens because, even before the transition, the citizen anticipates the values of the transition itself. By contrast, the ruler's preferred contract is non-extractive only after the transition. In this case, the ruler concedes the lowest tax cut that makes high effort incentive compatible for the citizen. In present value terms, at any $t < T^*$, the transition is therefore worth nothing to the citizen, whose value function is equal to the outside option.

At the moment of the transition from slow to fast growth, both contracts strictly benefit the citizen with respect to what would happen in an extractive equilibrium. This is also optimal for the ruler, who anticipates that, by reducing their share of the pie today, they will allow the pie to grow faster, and thus be able to have more pie in the future. In this sense, the ruler's patience is a key determinant of the transition.

To see this, consider the ruler's different consumption plans. In any period, along any contract, the ruler consumes all the output not left to the citizen, i.e., $x_{R,t} =$

$kA_t - x_{C,t} - c_h \mathbf{1}\{e_t = h\}$. Along σ^e, the citizen always exerts low effort, so the level of productivity in period t is $A_1 g_l^{t-1}$. The ruler's flow of consumption is $kA_1 g_l^{t-1} - \phi$. Along *any* RP social contract, the ruler's consumption exceeds this benchmark by

$$\underbrace{k\left(A_t - A_1 g_l^{t-1}\right)}_{\text{extra output}} - \underbrace{\left[x_{C,t} - \phi + c_h \mathbf{1}\{e_t = h\}\right]}_{\text{tax concession}}.$$

Focusing on σ_R^* and σ_C^*, we can see that before T^*, both terms are equal to zero. After T^*, both terms are positive.

Under σ_R^*, the tax concession is minimized and it is equal to the constant K (Remark 3), while the extra output term grows unboundedly as the state increases. Therefore, there exists a finite calendar period after which the latter term dominates the former. Under σ_C^*, both terms are instead linear in the state. The tax concession term increases linearly with constant slope $\frac{\delta k(g_h - g_l)}{1-\delta g_l}$ (see also below). The extra output term is also linear, but its slope increases over time.[12]

Therefore,

Remark 6. *Consider the two players' preferred contracts, i.e., σ_R^* and σ_C^*. Under both contracts, the ruler's consumption flow is lower than under the extractive contract for finitely many periods after T^*, and it is higher forever thereafter.*

Figure 6 illustrates this remark.

[12] To see this, write $A_t = A_{T^*} g_h^{t-T^*}$ and $A_1 g_l^{t-1} = A_{T^*} g_l^{t-T^*}$, so that

$$k\left(A_t - A_1 g_l^{t-1}\right) = kA_{T^*}\left(g_h^{t-T^*} - g_l^{t-T^*}\right) = kA_t\left[1 - (g_l/g_h)^{t-T^*}\right].$$

Its slope, $k\left[1 - (g_l/g_h)^{t-T^*}\right]$, is zero at T^* and increases towards k. It therefore eventually exceeds the slope of the tax concession term, which is lower than k because $\delta g_h < 1$.

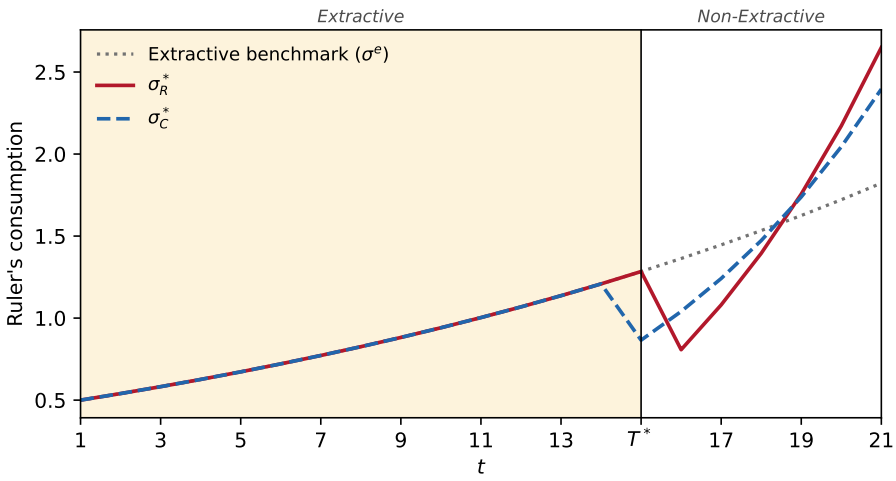


Figure 6: **The ruler's consumption flow along the two players' preferred contracts**.
Note: Parameters as in Figure 2, which imply $\tilde{A} > \hat{A}$, $A^* \approx 1.53$, and $T^* = 15$.

Another difference between the two contracts is that they deliver the tax concession to the citizen at different dates, and in different ways. Under σ_R^*, the ruler gives up a fixed *amount* of the pie in every period after the transition, and thus the citizen's rent is constant. As already emphasized, this amount is the lowest possible which makes it incentive compatible for the citizen to exert effort. By contrast, under σ_C^*, the ruler gives up a fixed *share* of output, $\frac{\delta(g_h - g_l)}{1-\delta g_l}$, and the citizen's rent, $\Delta(A)$, grows over time.

As a consequence, the comparison sharpens in the long run:

Remark 7. *As $A \to \infty$, the tax rate prescribed by σ_R^* converges to the extractive rate, $\tau^e(A)$. The tax rate prescribed by σ_C^* converges instead to $1 - \frac{\delta(g_h - g_l)}{1-\delta g_l} < 1$.*

6.2 Comparing the Two Contracts: Off-Path

Off-path, the two contracts serve as each other's punishment due to renegotiation proofness. If the ruler deviates in period t, the citizen exerts low effort in period t, and play follows σ_C^* from period $t+1$ onwards. If the citizen deviates, play follows σ_R^*. I now discuss why deviations are not appealing.

Consider the ruler first. Their most tempting deviation is to increase the tax rate once the citizen has already exerted high effort. How is that not optimal?

The punishment which deters such behavior is the loss of all future bargaining power. The citizen exerts low effort for one period, and play follows σ_C^* thereafter, which pushes the ruler down to the lower bound, $U_R^e(A)$. To see why, notice that, by deviating, the ruler collects at most $kA - \phi$ in tax revenues in the current period. They can never secure more than the whole output, net of the minimal payment that keeps the citizen in the game (see Section 4.2). Along the deviation, the ruler's continuation value is equal to $U_R^e(Ag_l)$, because the citizen exerts low effort for one period, and play then follows σ_C^*, which holds the ruler at their reservation value. Summing the two terms, $kA - \phi + \delta U_R^e(Ag_l) = U_R^e(A)$, so that the deviation can never be strictly profitable. In other words, the deviation turns the ruler into the extractive ruler of Section 4.2.[13]

This argument relies on how the citizen responds to the ruler's deviation:

Remark 8. *Consider the off-path prescriptions of the two players' preferred contracts, i.e., σ_R^* and σ_C^*. If the citizen keeps exerting high effort following a ruler's deviation, the deviation would be worth $kA-\phi+\delta U_R^e(Ag_h) > U_R^e(A)$, and it would be strictly profitable.*

Consider now the citizen's choice to cooperate. The most tempting deviation available to them is to exert low effort once the ruler has already announced a low tax rate. By doing so, the citizen increases their current period payoff. The punishment prescribes that the contract σ_R^* is in place starting the next period. This contract is restarted at a phase in which high effort has not been exerted, which pushes the citizen's continuation value down to their lower bound, $\bar{u}$. By deviating, the citizen thus loses the premium, $\omega - \bar{u} = \frac{c_h}{\delta}$, that the contract promises after every period of high effort. Since this loss exceeds the saved cost, c_h (as $\delta > 0$), exerting low effort is not profitable.

Before turning to the empirical predictions, I discuss the welfare properties of the model and its assumptions.

[13] This is shown formally in Step 2 of the proof of Lemma 3 (Appendix B).

7 Discussion on the Model

7.1 Welfare

Every RP social contract is second best (Lemma 4), meaning that no other equilibrium can deliver higher social surplus. In this section, I focus on the difference between second and first best contracts.

Let $\hat{T} \leq T^*$ denote the first calendar period at which the first best plan prescribes the citizen's high effort, i.e., the first period t such that $A_t \geq \hat{A}$. Let $n := T^* - \hat{T}$ denote the *delay* implied by incentive compatibility. The welfare loss, expressed in present value terms at $t = 1$, is the difference between the surplus of the first best plan and $\bar{\Pi}(A_1)$, i.e.,

$$L := \delta^{\hat{T}-1} \left[\Delta \left(A_{\hat{T}} \right) - \delta^n \Delta \left(A_{\hat{T}} g_l^n \right) \right] .$$

At $\hat{T}$, the social planner would collect all the surplus from the started path of sustained growth, $\Delta(A_{\hat{T}})$. Incentive compatibility forces society to wait n more periods. By then, productivity has grown and the surplus from sustained growth is larger, $\Delta(A_{\hat{T}} g_l^n) > \Delta(A_{\hat{T}})$, but it is discounted by δ^n. Since postponing high effort is suboptimal above $\hat{A}$, $L \geq 0$. Moreover, L is increasing in n, with $L = 0$ if and only if $n = 0$.

The loss is borne by the player who holds the bargaining power. Under σ_R^*, the surplus lost to the delay is entirely lost by the ruler. Under σ_C^*, the loss falls entirely on the citizen.

The delay, n, depends on the parameters of the model through the distance between the two thresholds,

$$\frac{\tilde{A}}{\hat{A}} = \frac{1 - \delta g_l}{\delta (1 - \delta) g_h} .$$

This ratio does not depend on c_h, k, or $\bar{u}$. It is decreasing in g_l, g_h and δ. In particular, when $\delta g_l + \delta (1 - \delta) g_h \geq 1$, i.e., when players are patient enough, there is no delay.

Patience lowers both $\tilde{A}$ and $\hat{A}$. The efficiency threshold, $\hat{A}$, falls with δ because a

more patient social planner weighs the return to the citizen's effort more heavily, and therefore starts it earlier. Indeed, as $\delta \to 0$, $\hat{A} \to \infty$. However, patience a further effect on $\tilde{A}$, as when players are more patient, inducing the citizen's effort is cheaper for the ruler.

7.2 Discussion of the Assumptions

In this subsection, I briefly comment on what assumptions are needed to obtain the main results of this paper.

Intertemporal Complementarity The ultimate driver of the dual transition is the intertemporal complementarity property of technological change. The return to the citizen's effort is proportional to the level of productivity, while its cost is not. Effort today raises productivity tomorrow, and thereby the return to effort tomorrow. As a result, high effort becomes worthwhile and rewardable, only once productivity is high enough.

To better see the role played by this assumption, suppose, by contrast, that returns to effort do not depend on the level of productivity, i.e., $A_{t+1} = g_l A_t + B\,\mathbf{1}\{e_t = h\}$, for some $B > 0$. The additional output generated by the citizen's effort is a constant. For this reason, the economy would be locked in a single regime from the start.

The Outside Option The baseline model assumes a constant outside option for the citizen. This assumption captures the idea that a citizen who does not accept the ruler's power loses access to capital as property rights are not enforced. However, one may also imagine that the size of the economy determines that of the outside option, for instance because revolts to overthrow rulers are more likely to succeed when societies are technologically advanced. One way to accommodate this intuition is by assuming that the outside option is increasing in the state, i.e., $\bar{u}(A) = uA$, for some small enough $u > 0$.

This alternative assumption has two main consequences. First, the extractive tax rate is a constant, $\tau^e = 1 - \frac{u(1-\delta g_l)}{k}$. Second, the minimal continuation value for the

citizen in a period of high effort, ω, grows with the state. By exerting low effort at state A, the citizen secures a continuation value of $\bar{u}(Ag_l)$, so that $\omega(A) = \bar{u}(Ag_l) + \frac{c_h}{\delta}$.

The main results, however, are robust to this alternative modeling choice. The social value of effort still increases with A, so the equilibrium still features a threshold rule. The transition, however, arrives earlier. To see why, recall that, to induce high effort, the ruler must promise the citizen a profit of $\frac{c_h}{\delta}$ on top of the value of the outside option, $\bar{u}$. The ruler can afford this promise only if the extra surplus generated by growth covers it, i.e., if $\Delta(Ag_h) \geq \frac{c_h}{\delta}$. This condition defines the threshold $\tilde{A}$. With a state-dependent outside option, the condition becomes $\Delta(Ag_h) \geq \frac{c_h}{\delta} - uA(g_h - g_l)$, which is easier to satisfy than in the baseline, so that $\tilde{A}$ decreases. The intuition is that effort also increases the citizen's future outside option, which the ruler must always concede. Part of the repayment for effort is thus paid through this channel, and promises are cheaper for the ruler.

The Identity of the Ruler The model features a single, infinitely-lived ruler. This is not strictly necessary for the results. The ruler is best understood as an *office*, rather than as a single person.

For instance, suppose that, in every period, the ruler is exogenously substituted with a new one with some probability, which then determines the discount factor δ.[14] A ruler who is substituted leaves the game and receives no further payoff. The new ruler inherits the history and the state A_t, and faces the same stage game as their predecessor.

The game is thus unchanged. Since a social contract is a profile of strategies defined on histories, and every ruler re-optimizes at every history, the equilibrium behavior of the office does not depend on who holds it. The sequence of rulers behaves as a single player. This observation matters for the empirical analysis of Section 8, where some dual transitions coincide with a regime change.

[14] To preserve a common discount factor, the citizen must be seen in the same way, e.g., as a dynasty.

8 Dual Transitions in History

The model makes four key testable predictions. First, the growth rate of the economy and the extractiveness of institutions change together. As period T^* is reached, growth accelerates and any social contract is non-extractive. Second, the ruler's share of output, τ_t, declines accordingly. Third, the ruler is *not* worse off. Their level of consumption, $x_{R,t} = \tau_t y_t$, eventually exceeds its level under the extractive contract. Fourth, political change is robust to renegotiation *because* it is sustained by growth.

In order to test these predictions, I collect data on countries' political institutions and income levels over the period 1789–2024. I proxy rulers' ability to expropriate resources from the economy using V-Dem's index of "legislative constraints on the executive" (Coppedge et al. [2025]), which records whether the legislature can question, investigate and oversee the executive. This index covers 199 countries from 1789 onwards. I obtain data on countries' GDP per capita from the Maddison Project Database (MPD, henceforth; Bolt and van Zanden [2025]).[15] Table 1 provides descriptive statistics.

	1789–1847	1848–1913	1914–1944	1945–2024	Total
A. Legislative constraints on the executive (0–1)					
Mean	0.30	0.41	0.41	0.49	0.44
Standard deviation	0.26	0.28	0.27	0.32	0.30
Countries w/ at least one obs.	63	112	119	182	199
B. GDP per capita (2011 international dollars)					
Mean	1,815	2,749	3,633	10,291	8,056
Median	1,580	2,156	2,578	5,255	3,612
Countries w/ at least one obs.	59	78	70	165	165

Table 1: **Summary statistics**.
Note: Each country-year pair is weighted equally.

8.1 Identifying Dual Transitions in the Data

In order to test the first prediction, I separately identify *political breaks* (i.e., sudden changes in the extractiveness of institutions) and *growth take-offs* (i.e., sustained increases in the growth rate of GDP per capita). I then ask whether growth take-offs are correlated with political breaks, as predicted by the model.

[15] For other uses of V-Dem data, see Funke et al. [2023] and O'Reilly and Murphy [2022]; for the MPD, see Acemoglu et al. [2005b] and Funke et al. [2023], among others.

I say that a country experiences a *political break* in year T if three conditions are satisfied: (i) on average, the index of legislative constraints is at least half a standard deviation (i.e., 0.15) higher in the decade starting in T (i.e., the period $[T, T+9]$) than in the decade before ($[T-10, T-1]$); (ii) during the decade starting in T, the index never falls below its average in the decade before; and (iii) in the following decade ($[T+10, T+19]$), the index is still at least a third of a standard deviation (0.10) above that average.[16]

The first condition identifies a large increase in the constraints faced by the executive, while the second and the third ensure that this increase is not temporary.

Analogously, I say that a country experiences a *growth take-off* in year T if four conditions are satisfied: (i) the average growth rate of GDP per capita is at least one percentage point higher in the ten years after T ($[T, T+10]$) than in the ten years before ($[T-10, T]$); (ii) in the ten years after T, GDP per capita grows by at least 1% a year on average; (iii) at the end of these ten years, GDP per capita is higher than in any year up to T; and (iv) in year T, GDP per capita is no more than 10% below its peak in the ten years before, and, after reaching that peak, it never fell more than 15% below it.[17]

The first two conditions identify a large acceleration of the growth rate, while the last two ensure that the acceleration is not simply capturing a recovery from a previous crisis. I exclude such rebounds because, in the model, dual transitions happen when productivity is high enough, not when output returns to a level already attained in the past. For instance, a recovery from a war or a pandemic accelerates the growth rate, but it is driven by the depth of the preceding recession rather than by productivity gains.

[16] Condition (iii) is checked only when the data cover $[T+10, T+19]$. Around a single reform, the three conditions can hold in several consecutive years. Consider the case of Portugal. The Carnation Revolution of April 1974 ended the Estado Novo dictatorship, and the index of legislative constraints rose from 0.03 in 1973 to 0.75 in 1974, and stabilized at levels above 0.9 from 1976 onwards. The three conditions then hold in every year from 1974 to 1982, because, for each of these years, the previous decade still includes years of dictatorship. To take care of this, I treat years less than ten years apart as a single political break, and date it in the year in which the increase in condition (i) is largest (i.e., 1974 in Portugal).

[17] For a similar approach to the identification of episodes of accelerations in GDP growth, see Hausmann et al. [2005]. Note also that I interpolate GDP per capita in missing years, assuming constant growth between two reported years across gaps of at most twenty-five years. As for political breaks, when multiple consecutive years satisfy these conditions, I date the take-off in the year in which the increase in condition (i) is largest.

A political break is a *dual transition* if a growth take-off is dated within a ten-year window. From 1789 to 2024, I identify 65 dual transitions in 58 different countries.[18] As a validation exercise, note that 61 of them coincide (within three years) with a regime change or a democratic transition recorded in at least one of four other datasets.[19] Appendix C presents statistics on their distribution over time and across countries.

Figure 7 shows the dynamics of GDP per capita around dual transitions. On average, the growth rate of GDP per capita rises from 1.2% a year in the decade before the political break to 3% a year in the decade after it.

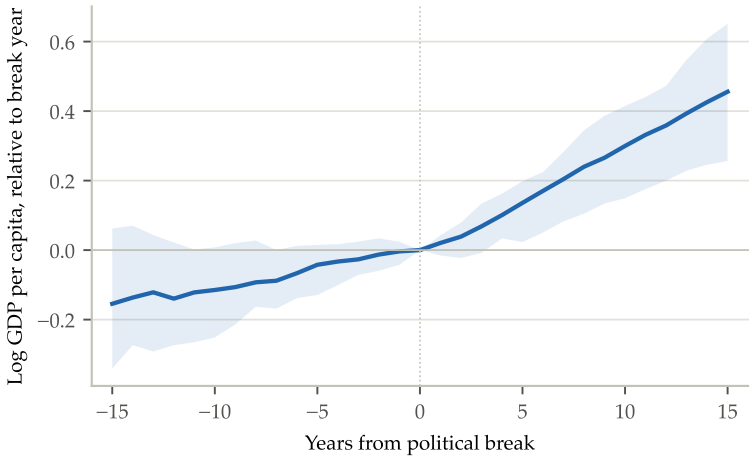


Figure 7: **GDP per capita around dual transitions**. Average of log GDP per capita, relative to its level in the year of the political break, by year from the break.
Note: All 65 dual transitions are included. The shaded band is the interquartile range across episodes (i.e., in each year, the 25th and 75th percentiles of the episodes' log GDP per capita relative to the break year).

The dual transition episodes include several well-known examples of political and economic modernization, under very different institutional settings. Among the cases in which a monarch conceded constraints on their power while keeping the throne, we find

[18] V-Dem follows each polity through changes in its borders, while the MPD uses present-day borders. Only two dual transitions occur within ten years of a country's split or a merger between countries (Czechia and Yemen, both in 1990).

[19] These are: (i) Polity5 (Marshall and Gurr [2020]), which scores regimes on an autocracy–democracy scale and separately measures constraints on the executive; (ii) the dataset of Boix et al. [2013], which classifies each regime as either a democracy or not; (iii) the Lexical Index of Electoral Democracy (Skaaning et al. [2015]), which ranks regimes by their electoral institutions; and (iv) V-Dem's Episodes of Regime Transformation (Maerz et al. [2024]), which dates periods of democratization and autocratization. I count a *match* between these sources and my classification if (i) records a regime change or a rise of at least two points in its score of executive constraints, if (ii) or (iii) records a transition to democracy, or if (iv) records the start of a democratization episode.

the Nordic kingdoms of Denmark and Norway. The Danish King Frederick VII renounced absolutism in 1848 and granted the June Constitution of 1849. Around this episode, the ten-year average growth rate of the economy rose from close to zero to about 2% a year. In Norway, the King accepted in 1884 that the government must answer to parliament. Growth took off in 1886, when the ten-year average growth rate of the economy rose from 0.2% to 2.1% a year.

In some other cases, dual transitions follow a regime change or a change in formal institutions. As discussed in Section 7.2, this is not necessarily in contrast with the model, whose predictions concern the office of the ruler, rather than the person holding it. France provides a famous example. After the fall of Napoleon III, the Constitutional Laws of 1875 established the parliamentary institutions of the Third Republic, and the ten-year average growth rate of the economy went from (near) zero to 1.7% a year.

For a complete list of all episodes of dual transitions, see the attached interactive Atlas.

One may be concerned that as political breaks and growth take-offs are both frequent, some would fall near each other *by chance*. To address this concern, I ask whether growth take-offs start more often in the decade after a political break than in other years. I estimate

$$TakeOff_{it} = \beta \, \mathbf{1}\left\{\text{political break in } [t-10, t]\right\}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{14}$$

where $TakeOff_{it}$ is a dummy equal to one if a growth take-off starts in country i in year t, and α_i and λ_t are country and year fixed effects, respectively (year fixed effects are included only in columns 2 and 4 of Table 2).

The parameter β is the difference between the probability that a take-off starts in the decade after a political break *versus* all other years. Table 2 reports its estimate in percentage points. It is positive and statistically significant in both specifications.

	Growth take-off starts in year t ($\times 100$)	
	(1)	(2)
Political break in $[t-10, t]$	1.40^{***}	1.05^{**}
	(0.50)	(0.47)
	$[p = 0.005]$	$[p = 0.028]$
Mean in comparison years	2.62	2.62
Country FE	Yes	Yes
Year FE	No	Yes
Growth take-offs	331	331
Country-years	12,061	12,061
Countries	162	162

Table 2: **Growth take-offs after political breaks**. The dependent variable is a dummy equal to one (times 100) if a growth take-off starts in year t.
Note: Standard errors, clustered by country, in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The effect is also *economically* significant. In the years that do not follow a political break (i.e., outside the decade after a break), a growth take-off starts with a probability of 2.62% per year (column 1). In the decade after a political break, this probability is 1.40 percentage points higher (about 50%). With year fixed effects (column 2), the excess probability is 1.05 percentage points (about 40%).

8.2 The Ruler's Share

In order to test the second prediction, I need a measure of the ruler's share of output, τ. I proxy for it using V-Dem's index of "executive embezzlement," which answers the question: "How often do members of the executive, or their agents, steal, embezzle, or misappropriate public funds or other state resources for personal or family use?" Experts answer this question and rate countries on a scale from 0 ("the executive treats all public resources as its own") to 4 ("never, or hardly ever"). I reverse and rescale this measure, so that the index goes from 0 (no embezzlement) to 1 (all public resources are embezzled).

To investigate how the embezzlement index evolves around political breaks, I estimate

$$\textit{Embezzlement}_{it} = \sum_{b \neq -5} \gamma_b \, \mathbf{1}\left\{\text{political break in } [t-b-4,\, t-b]\right\}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (15)$$

where $b \in \{-20, -15, \ldots, 25\}$ indexes five-year bins of years relative to the political break, the bin $b = -5$ (i.e., the five years before the break) is the reference, and α_i and λ_t are country and year fixed effects, respectively.

I estimate Equation (15) separately for dual transitions and for political breaks that are not followed by a growth take-off, each against the 91 countries that never had a political break. The coefficient γ_b measures how much embezzlement differs, in the years of bin b, from the five years before the break. Figure 8 plots the estimates. There is no evidence of pre-trends before dual transitions. After them, the index falls significantly and permanently.

After political breaks that are not followed by a growth take-off, the effect is never statistically different from zero. This result is consistent with the model. In an *empirically observed* dual transition, the ruler has two reasons to lower their share. First, their power is constrained. Second, as in the model, a smaller share of the pie is the price they pay to *sustain* the faster growth rate of the pie. When growth does not take off, only the first motive is present, and, even if rulers are more constrained, they do not reduce rent extraction.

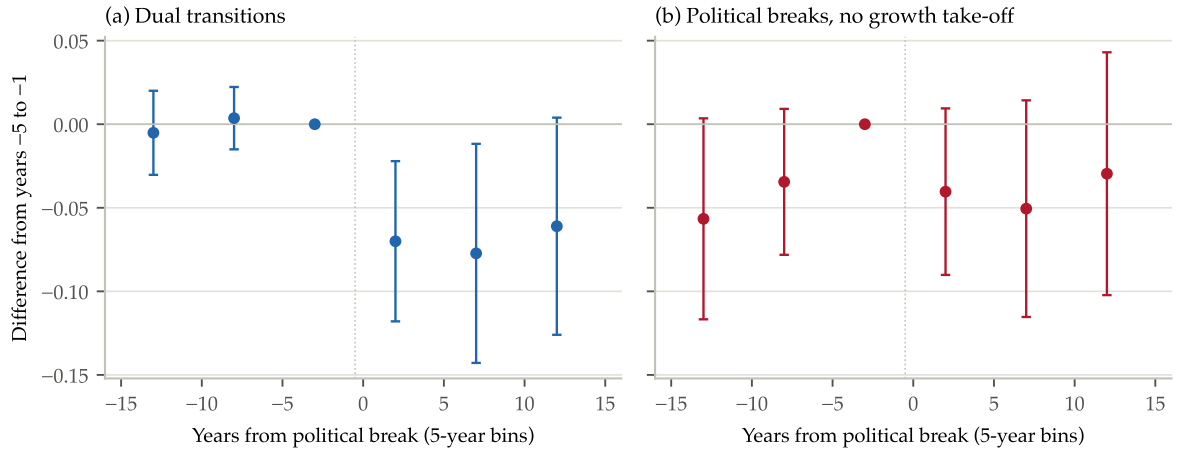


Figure 8: **Executive embezzlement around political breaks: Event Studies**. Coefficients γ_b of Equation (15), with 95% confidence intervals.
Note: Political breaks up to 1985, so that each episode can be observed for at least twenty-five years afterwards. When a country experiences more than one break of the same type, only the first is included (32 dual transitions, 22 political breaks without a growth take-off). Standard errors clustered by country.

8.3 The Ruler's Consumption

In order to test the third prediction, I compare the ruler's consumption, $x_R = \tau Y$, before and after each dual transition. Since the model's prediction concerns the *long run*

(Remark 6), I compare the ruler's average annual consumption over the twenty years after the political break with its average over the twenty years before it.

I measure output, Y, using data on countries' real GDP.[20] To measure the ruler's share, τ, I read the embezzlement index as the share of public resources that the executive treats as its own. This reading is not automatic, as experts answer in words and do not provide numbers or precise estimates. On the 0–1 scale, their five answers correspond to 0 (the executive embezzles "never, or hardly ever"), 0.25 ("selected others"), 0.5 ("about half"), 0.75 ("the rest"), and 1 ("all" public resources). I consider three alternative mappings from these values of the index to shares of output, which I illustrate in Figure 9. I also assume that public resources are a stable fraction of output, even if this assumption seems not decisive.[21]

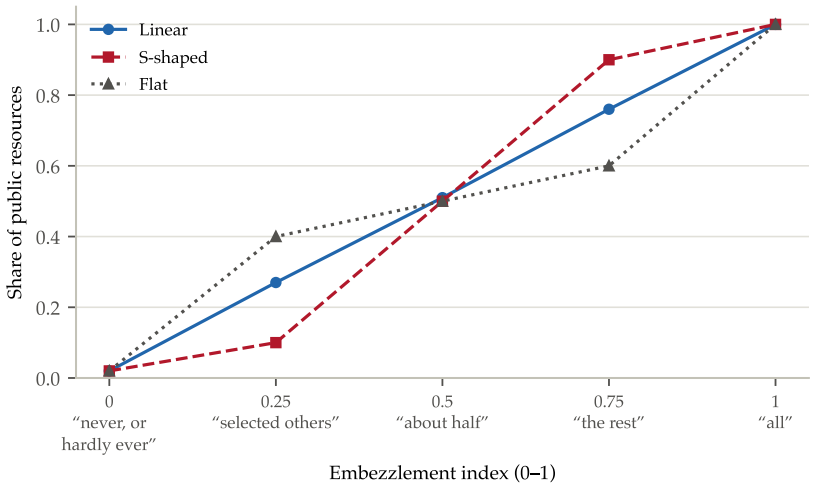


Figure 9: **From embezzlement answers to shares**. Share of public resources that the executive treats as its own, for each value of the embezzlement index that corresponds to an expert answer, under the three mappings.
Note: Between these values, the share is interpolated linearly.

In the median dual transition, average real GDP over the twenty years after the political break was 112% higher than over the twenty years before it. To offset this increase, the ruler's share would have had to fall by 53%. Instead, it fell by 8 to 16%,

[20] This is computed as GDP per capita from the MPD times population from Our World in Data.

[21] The ruler's consumption can be written as $x_R = s\,S\,Y$, where s is the share of public resources that the executive treats as its own and S is the ratio of public resources to output. The ruler's share of output is therefore $\tau = s\,S$. The test measures the change in s and holds S fixed. However, public revenues tend to increase relative to output after the executive is constrained (Dincecco [2009]). If so, the ruler's consumption after the transition is higher than I measure, and the results should be read as a lower bound.

depending on the mapping. Figure 10 shows, for each dual transition, the ruler's average consumption after the political break relative to its average before it.

Depending on the mapping, the ruler is not worse off in 48 to 53 of the 58 dual transitions for which the ruler's consumption can be measured over both windows, consistent with institutional change being a Pareto improvement, rather than a forced transfer from rulers to citizens. These results suggest that constitutions and other formal systems of checks and balances may then be a product of dual transitions, rather than their cause.

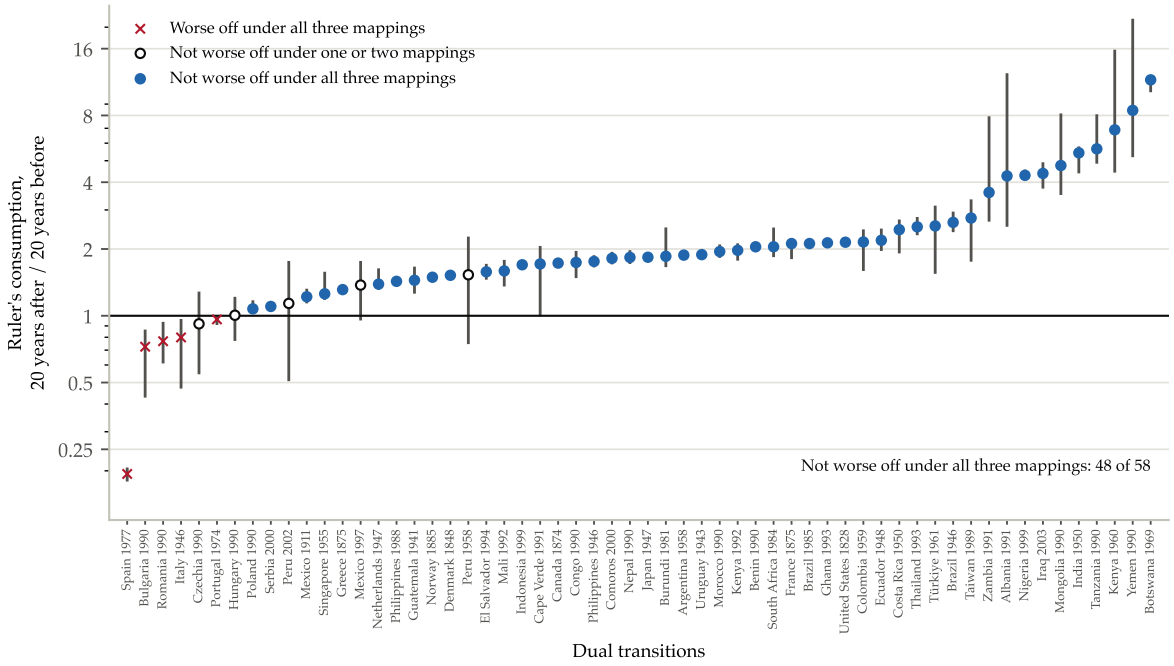


Figure 10: **The ruler's consumption after dual transitions**. Each bar corresponds to a dual transition, and shows the ruler's average consumption over the twenty years after the political break, divided by its average over the twenty years before it. The dot uses the linear mapping, and the bar shows the range across the three mappings.
Note: The vertical axis is in log scale. Labels below the bars indicate the country and the year of the political break.

Several historical episodes in the sample illustrate this logic. In Greece, Charilaos Trikoupis published in 1874 a newspaper article accusing King George I of appointing governments without parliamentary consent. After being jailed, Trikoupis was appointed prime minister only one year later, and the King committed to appoint only governments enjoying the "declared confidence of parliament" (the *Dedilomeni* principle). Even if no social uprising took place, the index of legislative constraints rose from 0.32 to 0.68, and the ten-year average growth rate of the economy rose from zero to 2.7% a year.

Norway provides another example. In 1884, the parliament forced the King's government out of office after a long constitutional dispute. Rather than resisting parliament, King Oscar II appointed the leader of the parliamentary majority as prime minister, establishing the principle that the government should reflect parliament's support. As a result, the index of legislative constraints rose from 0.74 to 0.93, and real GDP over the twenty years after the transition was about 50% higher than over the twenty years before.

In both cases, the ruler conceded constraints on their power without being overthrown, and growth took off within a decade, to the benefit of both rulers and citizens (see also Appendix C, Figure 13).

8.4 Renegotiation Robustness

In order to test the fourth prediction, I ask whether political changes that are not accompanied by a growth take-off are renegotiated more often than dual transitions.

Under the alternative (*uni-directional*) explanation according to which political change is the cause of growth, growth itself should have no impact on the likelihood that political change is renegotiated. By contrast, in my model, a concession made before period T^* cannot be renegotiation proof.

In this section, the persistence of political change is the outcome. Therefore, I date political breaks without condition (iii). This yields 125 breaks, 68 of which are dual transitions. I say that a political break is *reversed* in the first year in which the index of legislative constraints falls below its average in the decade before the break.[22]

Figure 11 plots the share of breaks not yet reversed, separately for dual transitions and for the other political breaks (Kaplan–Meier estimates). The two curves diverge substantially. After thirty years, 21% of dual transitions have been reversed, against 47% of the other breaks.

[22] In order not to count years that are already used to date the break and the take-off, I start measuring reversals in year $T + 20$. Breaks are followed until the end of the data and treated as *censored* when the data end.

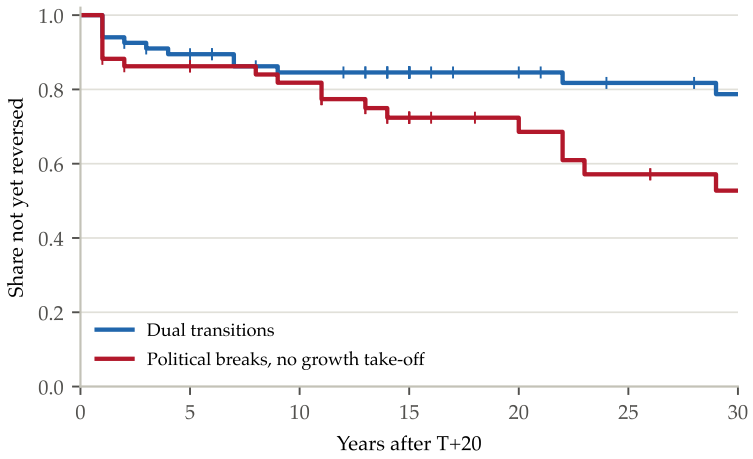


Figure 11: **Reversal of political breaks**. Share of political breaks whose constraints have not yet been reversed, by years from $T + 20$, where T is the year of the break (Kaplan–Meier estimates).
Note: Ticks mark breaks that leave the sample because the data end.

One may be concerned that this divergence result is simply driven by the fact that institutions in richer countries may be more stable regardless of growth accelerations. In order to address this concern, I estimate the following model.

For each political break i, let $p_i(s)$ be the probability that the break is reversed s years after $T + 20$, among the breaks that have not been reversed until then. I assume that

$$p_i(s) = p_0(s) \exp(\beta \, Dual_i + \gamma X_i), \tag{16}$$

where $p_0(s)$ is a probability common to all breaks, which is allowed to vary freely with s, $Dual_i$ is a dummy equal to one if the break is a dual transition, and X_i is a vector of controls measured in the year of the break.

By Equation (16), in any given year, the probability that a dual transition is reversed is e^{β} times the probability that a political break without a take-off is reversed, with the same controls. Table 3 reports estimates of e^{β} (the *hazard ratio*, in the language of survival analysis). A value below one means that dual transitions are reversed less often than other political breaks. Without controls (column 1), the probability of reversal after a dual transition is half that after other political breaks. Column 2 controls for log GDP per capita in the year of the break. The estimate does not change, and income itself does not predict reversals. Column 3 further controls for the *size* of the break (i.e., the

average of the index in the decade before the break and its gain), without significantly affecting the results.

	Probability of reversal after $T+20$		
	(1)	(2)	(3)
Dual transition	0.50**	0.50**	0.51*
	(0.17)	(0.17)	(0.18)
	[$p = 0.038$]	[$p = 0.039$]	[$p = 0.058$]
Log GDP per capita at the break		0.97	1.15
		(0.18)	(0.22)
		[$p = 0.884$]	[$p = 0.476$]
Size of the break	No	No	Yes
Reversals	35	35	35
Political breaks	118	118	118
Countries	96	96	96

Table 3: **Reversal of political breaks: dual transitions versus other breaks**. Hazard ratios from Cox proportional-hazards models of the time to reversal, measured from year $T + 20$.
Note: Standard errors, clustered by country, in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Consistent with the model, political changes that are followed by a growth take-off are robust over time, while those that are not are more often renegotiated.

9 Conclusions

This paper studies a simple society made of only two agents: a citizen and a ruler. In a simple repeated game, I show that slow growth and rent extraction cannot last forever. On the path of every Pareto efficient and renegotiation proof social contract, there comes a date at which the citizen starts to exert high effort, economic growth takes off, and the ruler concedes a non-extractive tax rate. Using cross-country data from 1789 onward, I present empirical evidence consistent with the main predictions of the model.

The historical reading of this result is that modern rulers may be less extractive than ancient ones not (only) because constitutions or other formal institutions constrain them, but (also) because citizens' work has become so productive that discouraging it through extraction is no longer in their own interest. In this view, constitutions are a product of the transition rather than its cause. The paper concludes by discussing descriptive empirical evidence consistent with this argument.

The model is simple and omits many interesting complications. For instance, I do not model political competition between multiple rulers and collective action problems among multiple citizens. These extensions are left for future research.

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A Proofs of the Main Results

Proof of Proposition 1. The proof proceeds in three steps.

- *Step 1: The strategy profile σ^e is an extractive social contract.*

  This proof (and all the subsequent ones) make use of the one-shot deviation principle, which applies because discounted payoffs are finite.

  Consider first the citizen's problem. Under σ^e, their continuation value is $\bar{u}$ in every period, so the potential gain from exerting high effort is $-c_h + \delta[\bar{u} - \bar{u}] = -c < 0$. Then, fixing the strategy of the ruler in period t, and the prescribed continuation play from $t+1$ onwards, the citizen has no incentives to exert high effort in period t. Consider now the citizen's participation choice. Fix a given tax rate τ. If $\tau \leq \tau^e(A)$, staying in the game yields at least $(1-\tau)kA + \delta\bar{u} \geq \phi + \delta\bar{u} = \bar{u}$. Then, staying is (weakly) optimal. If $\tau > \tau^e(A)$, staying yields strictly less than $\bar{u}$, and quitting the game is optimal.

  Consider now the ruler's problem. As shown above, under the prescribed continuation, the citizen always exerts low effort. Therefore, the tax rate chosen by the ruler in period t has no consequences on the level of productivity in period $t+1$, so that the ruler's problem is essentially static. Hence, $\tau^e(A)$ is optimal as it maximizes their current period payoff, subject to the binding participation constraint of the citizen.

  I now compute players' value functions. The citizen's value function is equal to $\bar{u}$ always. Then, the contract is always extractive on the equilibrium path. The ruler collects $\tau^e(A)kA = kA - \phi$ in taxes in every period. The growth rate of the economy is g_l. The ruler's value function is then equal to

$$U_R^e(A) = \sum_{t\geq 0} \delta^t \left(kAg_l^t - \phi\right) = \frac{kA}{1-\delta g_l} - \frac{\phi}{1-\delta} = \frac{kA}{1-\delta g_l} - \bar{u}\,,$$

  where the sum converges because $\delta g_l < \delta g_h < 1$.

- *Step 2: The citizen's lower bound.* In any period, the citizen can always quit the game and secure the outside option. Therefore, in any social contract, the citizen's value function must (weakly) exceed $\bar{u}$.

- *Step 3: The ruler's lower bound.* Fix any social contract σ with history H and state A. Fix $\varepsilon > 0$. Suppose the ruler announces $\tau^e(A) - \varepsilon$. By Step 2, the citizen's continuation value at any subsequent history is at least $\bar{u}$, so that staying in the game is strictly optimal: $\phi + \varepsilon kA + \delta\bar{u} > \bar{u}$. The ruler's consumption in the current period is $(1-\varepsilon)kA - \phi$. At the next period's history, H', the state is $A' \geq Ag_l$. Since σ is a social contract,

$$U_R(\sigma|H) \geq (1-\varepsilon)kA - \phi + \delta U_R(\sigma|H') .$$

  The same inequality holds at H', and at every history reached by repeating the deviation. Iterating T times,

$$U_R(\sigma|H) \geq \sum_{t=0}^{T-1} \delta^t \left[(1-\varepsilon)kAg_l^t - \phi\right] .$$

  Letting $T \to \infty$ and $\varepsilon \to 0$, I obtain $U_R(\sigma|H) \geq \frac{kA}{1-\delta g_l} - \frac{\phi}{1-\delta} = U_R^e(A)$.

□

Proof of Proposition 2. The proof proceeds in four steps. Step 1 defines the maximum equilibrium surplus, $\bar{\Pi}(A)$, and shows that it is attained by the threshold rule described in Section 5.2 (Lemma 2). Step 2 shows that every division of $\bar{\Pi}(A)$ which respects the two lower bounds of Proposition 1 is attainable by a social contract (Lemma 3). Consequently, Step 3 shows that a social contract is RP if and only if it attains $\bar{\Pi}(A)$ at every history (Lemma 4). Step 4 combines the previous steps to prove existence and the three statements shown in the Proposition. The proofs of all the auxiliary Lemmas are presented in Appendix B.

- *Step 1: The maximum equilibrium surplus.*

  I begin with the following definition.

  Definition 4 (Maximum Equilibrium Surplus)**.** *For any state A, let*

$$\bar{\Pi}(A) := \sup\left\{U_C(\sigma|H) + U_R(\sigma|H) \ : \ \sigma \text{ is a social contract, and } A(H) = A\right\}.$$

  The function $\bar{\Pi}(A)$ measures the largest total surplus that self-enforcing play can generate when the level of productivity is A.

  The next result characterizes $\bar{\Pi}(A)$, which is attained by the threshold rule described in Section 5.2.

  Lemma 2 (Characterization of Second Best Contracts)**.** *The maximum equilibrium surplus is*

$$\bar{\Pi}(A) = \begin{cases} \Pi_h(A) & \text{if } A \geq A^*, \\ \Pi_l(A) + \delta^{n(A)}\Delta\left(Ag_l^{n(A)}\right) & \text{if } A < A^*, \end{cases} \tag{17}$$

  where $n(A) := \min\{n : Ag_l^n \geq A^*\}$ is the number of periods until the state reaches the threshold A^*.

- *Step 2: The Pareto frontier is attainable.* The next Lemma shows that every division of the maximum equilibrium surplus which respects the lower bounds on players' payoffs (Proposition 1) is attainable in a social contract.

  Lemma 3 (The Pareto Frontier is Attainable)**.** *Fix any state $A > 0$, and any $w \in \left[\bar{u},\ \bar{\Pi}(A) - U_R^e(A)\right]$. There exists a social contract, σ_w, such that, at any history H with $A(H) = A$*

$$U_C(\sigma_w|H) = w, \qquad U_R(\sigma_w|H) = \bar{\Pi}(A) - w.$$

  Moreover,

$$U_C(\sigma_w|H') + U_R(\sigma_w|H') = \bar{\Pi}(A(H'))$$

at every subsequent (on- or off-path) history H'.

- *Step 3: Renegotiation proofness and second best efficiency.* Whenever a contract wastes surplus, one of the contracts constructed in Step 2 makes *both* players strictly better off. Since renegotiation proofness rules this out, any RP social contract is also second best from a social welfare perspective. Conversely, a contract that never wastes surplus leaves no room for Pareto efficient renegotiation:

  Lemma 4 (Renegotiation Proofness is Second Best Efficiency)**.** *Fix any state $A > 0$. A social contract σ belongs to $\Sigma(A)$ if and only if, at every (on- or off-path) history H, $U_C(\sigma|H) + U_R(\sigma|H) = \bar{\Pi}(A(H))$.*

- *Step 4: Existence and equilibrium properties.* Existence follows from the previous two steps. By Lemma 3, σ_w attains $\bar{\Pi}(A(H))$ at every history. Then, by Lemma 4, $\sigma_w \in \Sigma(A_1)$, so that $\Sigma(A_1) \neq \emptyset$.

  Now fix any $\sigma \in \Sigma(A_1)$. I prove the three statements in turn.

  Statement (i). By Lemma 4, σ attains $\bar{\Pi}(A(H))$ at every on-path history, so it prescribes the threshold rule of Lemma 2. The threshold rule is indeed the unique plan attaining $\bar{\Pi}(A)$. The citizen's effort when the state is below A^* is either not incentive compatible for them or suboptimal for society (Step 1 of the proof of Lemma 2). In addition, prescribing low effort at any state above $\hat{A}$ strictly reduces surplus (see Footnote 9).

  Statement (ii). Consider first the floors. The floor $\bar{u}$ follows directly from Proposition 1. Moreover, every period $t > T^*$ follows a period of high effort (Statement (i)), so $U_C(\sigma|H_t) \geq \omega$ by Equation (8).

  For the ceilings, I use the ruler's lower bound. By Lemma 4, $U_C(\sigma|H_t) + U_R(\sigma|H_t) = \bar{\Pi}(A_t)$, and, by Proposition 1, $U_R(\sigma|H_t) \geq U_R^e(A_t) = \Pi_l(A_t) - \bar{u}$. Therefore,

$$U_C(\sigma|H_t) \;\leq\; \bar{u} + \left[\bar{\Pi}(A_t) - \Pi_l(A_t)\right].$$

If $t > T^*$, then $A_t \geq A^*$, so that $\bar{\Pi}(A_t) = \Pi_h(A_t)$ and the term in bracket is equal to $\Delta(A_t)$.

If $t \leq T^*$, then $n(A_t) = T^* - t$ and $A_t g_l^{T^*-t} = A_{T^*}$, so that, by Equation (17), the term in bracket is equal to $\delta^{T^*-t}\Delta(A_{T^*})$. This proves Equation (12). Finally, Equation (13) is simply the citizen's Bellman equation.

Statement (iii). By Statement (ii) and the definition of ω, at every $t > T^*$, $U_C(\sigma|H_t) \geq \omega = \bar{u} + \frac{c_h}{\delta} > \bar{u}$. Then, the contract is non-extractive at every on-path history with $t > T^*$. Finally, iterating Equation (13),

$$\sum_{s\geq 0} \delta^s (1 - \tau_{t+s}) kA_{t+s} = U_C(\sigma|H_t) + \frac{c_h}{1-\delta} > \bar{u} = \sum_{s\geq 0} \delta^s (1 - \tau^e(A_{t+s})) kA_{t+s}.$$

Hence, $\tau_{t+s} < \tau^e(A_{t+s})$ at some $s \geq 0$. This completes the proof.

□

B Proofs of the Auxiliary Results

This appendix collects the proofs of Lemma 1 (Section 5) and of the three auxiliary Lemmas stated in the Proof of Proposition 2 (Appendix A).

Proof of Lemma 1. For any plan $\{e_t\}_{t=1}^{\infty}$, society's surplus can be written as

$$\sum_{t\geq 0} \delta^t kA \prod_{s<t} g_{e_s} \; - \; \sum_{t\geq 0} \delta^t c_h \mathbf{1}\{e_t = h\} \; = \; kA \cdot G \; - \; C, \tag{18}$$

where $G := \sum_{t\geq 0} \delta^t \prod_{s<t} g_{e_s}$ and $C := \sum_{t\geq 0} \delta^t c_h \mathbf{1}\{e_t = h\}$ depend on the plan but not on the state.

Recall that $F(A)$ denotes the value function of the social planner problem (Section 5). It is the supremum of the family of linear increasing functions defined by Equation (18). Therefore, it is increasing and convex. Moreover, it solves the following Bellman

equation:

$$F(A) = kA + \max\left\{\delta F(Ag_l),\ -c_h + \delta F(Ag_h)\right\}.$$

High effort is optimal at state A if and only if

$$\delta\left[F(Ag_h) - F(Ag_l)\right] \geq c_h. \tag{19}$$

Since F is increasing and convex, if Condition (19) holds at A, it holds at every $A' \geq A$. □

Proof of Lemma 2. The proof follows in two steps.

- *Step 1: No social contract generates more surplus than the threshold rule.* Recall that the function $F(A)$ is the value function of the social planner problem (see the proof of Lemma 1). Then, by definition, $\bar{\Pi}(A) \leq F(A)$. The first best plan (see Section 5) prescribes low effort below $\hat{A}$, and high effort forever thereafter. Hence, $F(A') = \Pi_h(A')$ at any state $A' \geq \hat{A}$, while at any state $A' < \hat{A}$, $F(A') - \Pi_l(A') = \delta^m \Delta(A' g_l^m)$, where m is the number of periods until the state reaches $\hat{A}$.

  I distinguish two cases.

  - If $\tilde{A} \leq \hat{A}$, then $A^* = \hat{A}$, so the threshold rule coincides with the first best plan and its value is exactly $F(A)$. In this case, the first best and the second best plans coincide, so the claim follows immediately.
  - Suppose instead $\tilde{A} > \hat{A}$, so that $A^* = \tilde{A}$. By Condition (9), if high effort at state A is followed by high effort forever, incentive compatibility requires $A \geq \tilde{A}$. I now show that $A \geq \tilde{A}$ is necessary for the citizen to exert high effort in *any* social contract, whatever the continuation.

    Consider any social contract which prescribes high effort at some history with state A. In the next period, the state is Ag_h. The citizen must receive a continuation value of at least ω (Equation (8)), and the ruler a continuation

value of at least $U_R^e(Ag_h)$ (Proposition 1). Since continuation play is itself a social contract, the two continuation values sum to at most $F(Ag_h)$. Using $\omega = \bar{u} + \frac{c_h}{\delta}$ and $U_R^e(Ag_h) = \Pi_l(Ag_h) - \bar{u}$, high effort at state A therefore requires

$$F(Ag_h) - \Pi_l(Ag_h) \geq \frac{c_h}{\delta}. \tag{20}$$

Condition (20) fails at every $A < \tilde{A}$.[23] Hence, no social contract prescribes high effort below A^*.

It follows that no social contract generates more surplus than the threshold rule ("exert high effort if and only if the state is at least A^*"). It remains to show that the threshold rule can be implemented in a social contract.

- *Step 2: The threshold rule can be implemented in a social contract.* Consider the following profile of strategies. The citizen exerts high effort if and only if $A \geq A^*$ (and the ruler has not deviated in the current period), and stays in the game if and only if $\tau \leq \tau^e(A)$. The ruler sets $\tau = \tau^e(A)$ up to and including the first high-effort period, and $\tau = 1 - \frac{\phi + c_h/\delta}{kA}$ afterwards. Any deviation triggers permanent reversion to σ^e, which is a social contract by Proposition 1.

  The citizen's continuation value is $\bar{u}$ at every history up to and including the first high-effort period, and ω afterwards. Then, they have no strictly profitable deviation. Consider now the ruler. At any history, their most profitable deviation is to raise the tax to $\tau^e(A)$. Under the prescribed on-path continuation, any higher rate triggers the citizen's exit. This deviation (which can possibly be profitable only at $A \geq A^*$) would allow the ruler to collect $kA - \phi$ in taxes today. Since any deviation is followed by σ^e, the deviation is worth $kA - \phi + \delta U_R^e(Ag_l) = U_R^e(A)$.

[23] The LHS of Condition (20) is the surplus the first best plan generates from state Ag_h in excess of permanent low effort. There are two cases. If $Ag_h \geq \hat{A}$, the first best plan prescribes high effort forever from Ag_h, so that $F(Ag_h) = \Pi_h(Ag_h)$ and the LHS is equal to $\Delta(Ag_h)$. Condition (20) then reduces to Condition (9), which fails when $A < \tilde{A}$. If $Ag_h < \hat{A}$, the first best plan prescribes low effort until the state reaches $\hat{A}$, and high effort forever thereafter. The LHS is then equal to $\delta^m \Delta(Ag_h g_l^m)$, where m is the number of periods until the state reaches $\hat{A}$ (see the opening of Step 1). Then, $Ag_h g_l^{m-1} < \hat{A}$, which implies $Ag_h g_l^m < \hat{A} g_l$. Hence, the LHS is bounded above by $\Delta\left(\hat{A} g_l\right)$. Finally, $\Delta\left(\hat{A} g_l\right) < \Delta\left(\tilde{A} g_h\right) = \frac{c_h}{\delta}$, so that Condition (20) fails whenever the state is below $\tilde{A}$.

Therefore, the ruler does not have a strictly profitable deviation if their value function (weakly) exceeds $U_R^e(A)$ at every on-path history.

I verify this condition by considering two different cases. First, at any $A \geq A^*$, the ruler's value function is equal to the total surplus net of the citizen's value function. Then, using $U_R^e(A) = \Pi_l(A) - \bar{u}$, the constraint reduces to

$$\underbrace{\Pi_h(A) - \bar{u} \geq U_R^e(A)}_{\text{At the first high-effort period}} \iff \Delta(A) \geq 0;$$

$$\underbrace{\Pi_h(A) - \omega \geq U_R^e(A)}_{\text{At later high-effort periods}} \iff \Delta(A) \geq \frac{c_h}{\delta}.$$

Notice that both conditions follow from the definition of A^*.[24]

Second, at any $A < A^*$, the ruler anticipates the future stream of payoffs which will start materializing after $n(A)$ periods. Therefore, their value function is (weakly) larger than their lower bound, $U_R^e(A)$.

By Step 1, no social contract generates more surplus than the threshold rule. By Step 2, the threshold rule can be implemented as a social contract. Hence, $\bar{\Pi}(A)$ equals the present value of the threshold rule. If $A \geq A^*$, high effort starts immediately, so this value is $\Pi_h(A)$. If $A < A^*$, the threshold rule coincides with permanent low effort for $n(A)$ periods, and then switches to permanent high effort at state $Ag_l^{n(A)}$. Relative to $\Pi_l(A)$, its value therefore adds $\delta^{n(A)}\Delta\left(Ag_l^{n(A)}\right)$. This is Equation (17).

□

Proof of Lemma 3. The proof proceeds in three steps. Step 1 constructs two *extreme* contracts: the *ruler-preferred* contract, σ_R^*, and the *citizen-preferred* contract, σ_C^*. Step

[24] In particular, the first condition follows from noticing that $A \geq A^* \geq \hat{A}$, and $\Delta\left(\hat{A}\right) = F\left(\hat{A}\right) - \Pi_l\left(\hat{A}\right) \geq 0$ by definition. For the second, notice that $A \geq A^* g_h \geq \tilde{A} g_h$, and $\Delta\left(\tilde{A} g_h\right) = \frac{c_h}{\delta}$ by definition of $\tilde{A}$.

2 verifies that both profiles are social contracts. Step 3 constructs the intermediate contracts σ_w by mixing over the tax rates of the two extreme contracts.

- *Step 1: The two extreme contracts.* Both contracts prescribe the citizen to exert high effort if and only if $A \geq A^*$, and to stay in the game if and only if $\tau \leq \tau^e(A)$. They differ only in how taxes split the surplus generated by growth.

  Moreover, the two contracts serve as each other's punishments. If the ruler deviates from the prescribed tax rate (which is different depending on the contract), play follows σ_C^* from the next period onwards. If instead the citizen deviates, play follows σ_R^* from the next period onwards. The citizen deviates when exerting low effort at a history where high effort is prescribed, when exerting high effort in a period in which the ruler deviated, or when staying in the game after a tax rate above $\tau^e(A)$.

  The ruler-preferred contract. Under σ_R^*, the citizen's value function is equal to its lower bound (which depends on their effort level; see Proposition 1 and Equation (8)), and the ruler receives the remainder of the surplus:

$$U_C(\sigma_R^*|H_t) = \begin{cases} \bar{u} & \text{for } t \leq T^*, \\ \omega & \text{for } t > T^*, \end{cases} \qquad U_R(\sigma_R^*|H_t) = \bar{\Pi}(A_t) - U_C(\sigma_R^*|H_t).$$

  The equilibrium tax rates follow. Up to (and including) the period of the transition from slow to fast growth, the citizen's value function is constant at $\bar{u}$, so their flow payoff is ϕ and the tax rate is exactly $\tau^e(A)$. After the transition, their value function is equal to the constant ω, so their flow payoff is the constant $(1-\delta)\omega + c_h = \phi + c_h/\delta$. The ruler-preferred contract, σ_R^*, therefore prescribes[25]

$$\tau_t = \tau^e(A_t) \quad \text{for } t \leq T^*, \qquad \tau_t = 1 - \frac{\phi + c_h/\delta}{kA_t} \quad \text{for } t > T^*.$$

[25] Assuming that $kA^*\left(1 - \frac{\delta(g_h - g_l)}{1-\delta g_l}\right) \geq \phi$ and $kA^* g_h \geq \phi + c_h/\delta$, keeps both contracts' tax rates in $[0,1]$ at every reachable state.

The citizen-preferred contract. Under σ_C^*, the ruler's value function is equal to its lower bound (Proposition 1), and the citizen receives the remainder of the surplus:

$$U_R\left(\sigma_C^*|H_t\right) = U_R^e\left(A_t\right) , \qquad U_C\left(\sigma_C^*|H_t\right) = \bar{\Pi}\left(A_t\right) - U_R^e\left(A_t\right) .$$

The equilibrium tax rates follow. Before the transition ($A < A^*$), the citizen's value function is $\bar{u} + \delta^n \Delta\left(A_{T^*}\right)$, where n is the number of periods before T^*. Their flow payoff is then $\left(\bar{u} + \delta^n \Delta\right) - \delta\left(\bar{u} + \delta^{n-1}\Delta\right) = \phi$. After the transition ($A \geq A^*$), the citizen's flow payoff is $\left[\bar{u} + \Delta\left(A\right)\right] + c_h - \delta\left[\bar{u} + \Delta\left(Ag_h\right)\right] = \phi + \frac{\delta k(g_h - g_l)}{1-\delta g_l} A$, using the fact that Δ is linear in the state. The citizen-preferred contract, σ_C^*, therefore prescribes

$$\tau_t = \tau^e\left(A_t\right) \quad \text{for } t < T^* , \qquad\qquad \tau_t = \tau^e\left(A_t\right) - \frac{\delta\left(g_h - g_l\right)}{1 - \delta g_l} \quad \text{for } t \geq T^* .$$

- *Step 2: Both profiles are social contracts.* Because each contract serves as the other's punishment, their equilibrium properties are verified jointly.

  - *The citizen has no strictly profitable deviations.* Consider first their choice of effort, at a history where high effort is prescribed by the candidate social contract. The current period's flow payoff is the same regardless of the effort choice. Complying with the contract costs c_h today, and leads to either σ_R^* or σ_C^* at state Ag_h, which implies a continuation value of at least ω. Low effort costs nothing today, but leads to a restarted σ_R^* contract, at state Ag_l, which is worth $\bar{u}$.[26] Then, complying to the contract is always (weakly) optimal. Finally, since both contracts leave the citizen with a value function of at least $\bar{u}$ in every period, quitting the game is never optimal.

  - *The ruler has no strictly profitable deviation.* At any history, the ruler's most

[26] The two branches of U_C in Step 1 are indexed by calendar time because, on-path, high effort begins at T^*. However, what σ_R^* actually tracks is whether high effort has been exerted by the citizen in the previous period. Therefore, a punishment restarts σ_R^*, at a phase in which the citizen had exerted low effort, so it is worth $\bar{u}$ at any state — including states reached after T^*. The same applies to the citizen's deviations under σ_C^*, considered next.

profitable deviation is to raise the tax rate to $\tau^e(A)$. Given the prescribed play, any higher rate triggers the citizen's exit.[27] This deviation would allow the ruler to collect $kA - \phi$ in taxes today.

How would the citizen respond to such deviation? They never find it (strictly) optimal to leave the game. Consider their choice of effort.[28] Exerting high effort in a period in which the ruler deviated is itself a deviation, and therefore leads to σ_R^*, which yields $-c_h + \delta\omega = \delta\bar{u}$. By contrast, exerting low effort leads to σ_C^*, and yields $\delta\left[\bar{u} + \bar{\Pi}(Ag_l) - \Pi_l(Ag_l)\right] \geq \delta\bar{u}$. Then, the ruler's deviation forces the citizen to exert low effort for one period, followed by σ_C^* forever thereafter. For the ruler, this is worth $kA - \phi + \delta U_R^e(Ag_l) = U_R^e(A)$. Then, it is never (strictly) profitable.

- *Step 3: The intermediate contracts.* Fix some $w \in \left[\bar{u},\ \bar{\Pi}(A) - U_R^e(A)\right]$. There exists $\lambda \in [0,1]$ such that $w = \lambda\left[\bar{\Pi}(A) - U_R^e(A)\right] + (1-\lambda)\bar{u}$.

  The candidate contract, σ_w, prescribes the same actions and the same punishments as the two extreme contracts. The citizen exerts high effort if and only if $A \geq A^*$, stays in the game if and only if $\tau \leq \tau^e(A)$, and exerts low effort in any period in which the ruler deviated. On-path, in every period, the ruler sets the λ-weighted average of the tax rates prescribed by σ_C^* and σ_R^*. Off-path, if the ruler deviates, play follows σ_C^*; if the citizen deviates, play follows σ_R^*.

  Three observations complete the proof.

  First, *feasibility*. In every period, the prescribed tax rate is a convex combination of the two extreme rates. Under the maintained parametric assumptions (see Footnote 25), both lie in $[0, \tau^e(A)]$; hence, so does their average.

  Second, *second best efficiency*. On-path, the three contracts prescribe identical actions, and differ only in the equilibrium tax rates, which are pure transfers.

[27] Quitting is optimal for the citizen at that history. Staying would be a deviation, after which play follows σ_R^*, worth $\bar{u}$ to the citizen. Staying therefore yields $(1-\tau)kA + \delta\bar{u} < \phi + \delta\bar{u} = \bar{u}$, since $\tau > \tau^e(A)$.

[28] This response matters because it determines how fast the state grows in the deviation period, and hence how much the deviation is worth to the ruler.

Since the citizen's flow payoff is linear in the tax rate, at every on-path history their continuation value under σ_w is the λ-average of their values under σ_C^* and σ_R^*. Since total surplus is always maximized and equal to $\bar{\Pi}(A)$, the ruler receives the remainder. In particular, the citizen's value at the initial state is w, and the ruler's is $\bar{\Pi}(A) - w$. At every off-path history, continuation play is a restarted σ_R^* or σ_C^*, whose value functions also sum to $\bar{\Pi}(A(H))$ (Step 1). Hence, the two value functions sum to $\bar{\Pi}(A(H))$ at every history.

Third, *incentive compatibility*. The checks on players' incentive compatibility constraints follow directly from those in Step 2.

□

Proof of Lemma 4. Fix a social contract σ and a history H. By Definition 4, no social contract of the continuation game can deliver more than $\bar{\Pi}(A(H))$ in total.

(*If.*) If σ attains $\bar{\Pi}(A(H))$ at every history H, no alternative contract can make both players strictly better off. By Definition 3, σ is renegotiation proof.

(*Only if.*) Suppose instead that σ wastes surplus at some H. Give the citizen slightly more than $U_C(\sigma|H)$, and the ruler the rest of $\bar{\Pi}(A(H))$. Both payoffs respect the lower bounds of Proposition 1, so, by Lemma 3, this division is attainable by a social contract of the continuation game. It makes both players strictly better off, so σ is not renegotiation proof. □

C Dual Transitions: Descriptive Statistics

This appendix collects additional material for Section 8. It shows where and when dual transitions happened (Table 4, Figure 12), and how *violent* they were (Figure 13).

	1789–1847	1848–1913	1914–1944	1945–2024	Total
Europe & C. Asia	0/2	4/8	1/1	14/18	19/29
North America	1/1	1/1	0/0	0/0	2/2
Latin America	0/0	1/4	2/3	11/19	14/26
East Asia & Pacific	0/0	0/1	0/1	8/13	8/15
South Asia	0/0	0/0	0/1	2/3	2/4
Middle East & N. Africa	0/0	0/0	0/0	3/7	3/7
Sub-Saharan Africa	0/0	0/0	1/2	16/32	17/34
All	1/3	6/14	4/8	54/92	65/117

Table 4: **Dual transitions by region and era**. Each cell reports the number of dual transitions over the number of political breaks.
Note: Only political breaks with income data around their date are counted. Regions follow the World Bank classification.

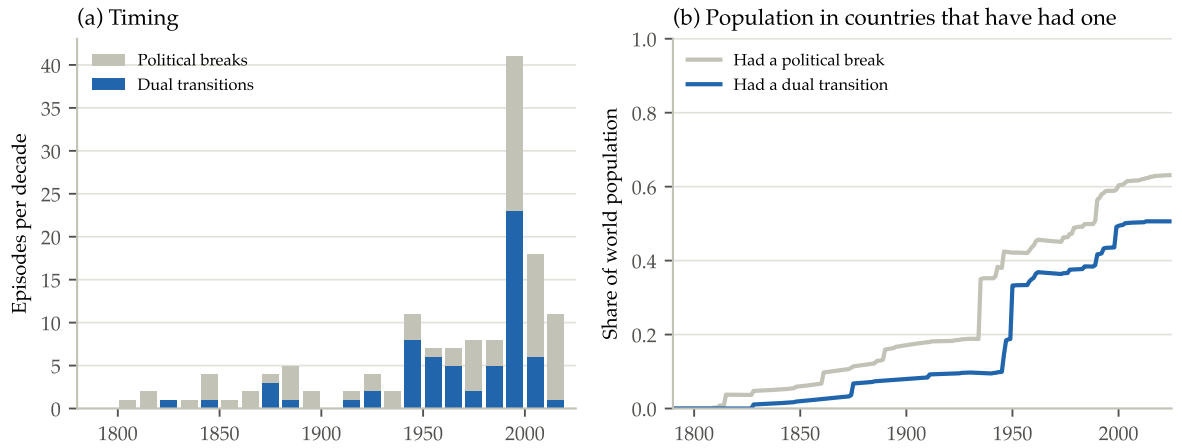


Figure 12: **Dual transitions over time**.
Note: Political breaks after about 2013 cannot be dual transitions, because a growth take-off can only be identified with ten years of income data after it. Population data are from Our World in Data.

Violence The model predicts that rulers concede constraints on their power voluntarily, rather than under the threat of force. To assess whether this is the case in the data, I ask how each of the 65 dual transitions happened.

I consider the events that took place in the three years before and after the political break (i.e., the window $[T-3, T+3]$, where T is the year of the break), and that are connected to the political break.

For each dual transition, I code two binary variables. The first dummy variable records whether the ruler was *removed by force*.[29] The second dummy variable records

[29] Consistent with the rest of the paper, the ruler is defined as whoever holds executive power before the political break, i.e., the monarch, or the dynasty, in a monarchy; the dictator, or the ruling party in an autocracy; the colonial power in a colony.

whether the transition involved *large-scale violence*, i.e., a revolution, a civil war, or a war, with roughly 1,000 political deaths or more. Figure 13 shows how the dual transitions are distributed across the four combinations of these two variables, by decade.

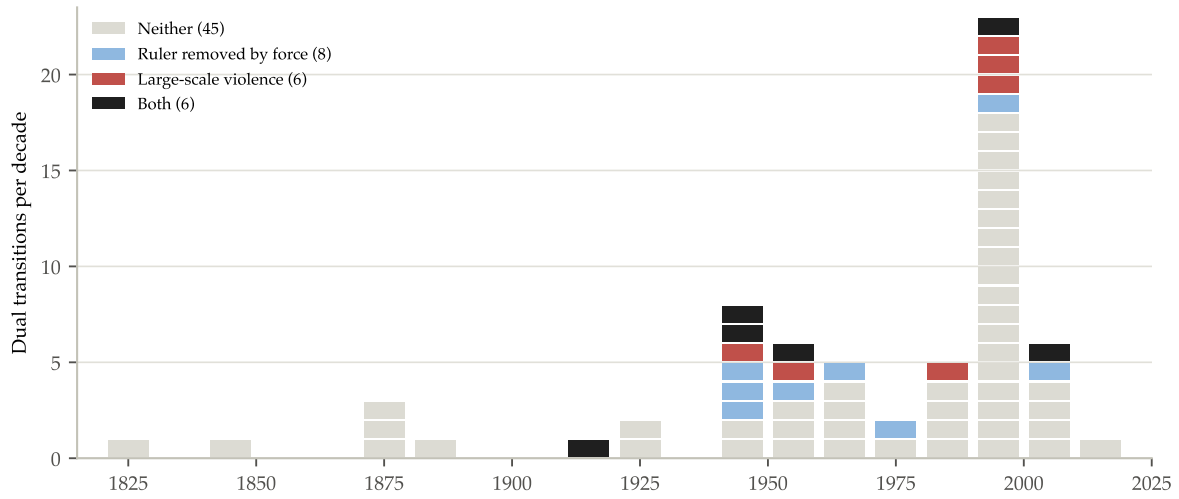


Figure 13: **Violence in dual transitions**. Each square is a dual transition. Large-scale violence means roughly 1,000 political deaths or more.
Note: Events in the three years before and after the political break, connected to the break itself.

The coding is performed by a large language model (Claude Opus 5.5) with access to web search. For each dual transition, the model receives the country, the year of the political break, and a fixed written prompt describing the window, the definition of the ruler, and the two variables above. The model is then instructed to verify the relevant events against historical sources, and to return the two variables together with a brief account of the events, the sources it relied on, and a self-assessed level of confidence.[30]

[30]The model never reported a low level of confidence.